Work Organization and Cumulative Advantage

by

Dylan Nelson
UIUC Gies

Nathan Wilmers
MIT Sloan

Letian Zhang
HBS

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Abstract

Over decades of wage stagnation, researchers have argued that reorganizing work can boost pay for disadvantaged workers. But upgrading jobs could inadvertently shift hiring away from those workers, exacerbating their disadvantage. We theorize how work organization affects cumulative advantage in the labor market, or the extent to which high-paying positions are increasingly allocated to already-advantaged workers. Specifically, raising technical skill demands exacerbates cumulative advantage by shifting hiring towards higher-skilled applicants. In contrast, when employers increase autonomy or skills learned on-the-job, they raise wages to buy worker consent or commitment, rather than pre-existing skill. To test this idea, we match administrative earnings to task descriptions from job posts. We compare earnings for workers hired into the same occupation and firm, but under different task allocations. When employers raise complexity and autonomy, new hires' starting earnings increase and grow faster. However, while the earnings boost from complex, technical tasks shifts employment toward workers with higher prior earnings, worker selection changes less for tasks learned on-the-job and very little for high autonomy tasks. These results demonstrate how reorganizing work can interrupt cumulative advantage.

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1 Introduction

Employers typically allocate high-paying jobs to already-advantaged workers. High-paying restaurants hire more experienced waitstaff; high-wage retailer Trader Joe's is pickier in hiring its cashiers than Dollar General is for its minimum wage equivalents. As a result, workers with higher-paying, more impressive prior work experience can monopolize subsequent high-paying jobs. What forms of work organization might interrupt this source of cumulative advantage in the labor market?

Researchers have investigated a number of alternative approaches to work organization. Good jobs strategies (Ton, 2014), lean production (Vidal, 2022), and high-involvement work systems (Osterman, 2006) all seek to engage and empower workers in support of higher productivity and pay. Yet many organizational changes that raise pay also shunt jobs to workers who have higher education, more experience, or other sources of individual advantage. Increasing pay, in these cases, may actually *exacerbate* cumulative advantage in the labor market. Indeed, Trader Joe's, with a hiring process that, relative to competitors, favors higher-educated and more experienced workers, is often held up as the model of multi-tasking and frontline retail job design. More broadly, high-involvement work systems pay production worker more, but also employ more educated workers (Osterman, 2006; Caroli and Van Reenen, 2001). Companies that embrace structured management practices, like continuous improvement and shared production targets, show similar patterns of worker skill selection (Bender et al., 2018). Even the unionized manufacturing jobs of post-World War II America were the core to a balkanized, dual labor market that consigned marginalized workers into peripheral jobs (Edwards et al., 1975).

These examples demonstrate a dilemma for theories that link work organization to broader issues of inequality and exclusion: An employer that upgrades an entry-level job and raises its wage may simultaneously substitute in a more experienced worker, thereby undermining rather than improving opportunity. Trader Joe's may pay more to cashiers than Dollar General, but if it disproportionately employs more highly educated workers,

then it could actually consolidate positions of privilege for relatively advantaged workers (Wilmers and Aeppli, 2021).

This dilemma in turn reflects a fundamental organizational selection process: higher wage jobs attract higher-performing workers. As long as employers prefer higher-skill workers to lower-skill workers at a given wage, then even an externally-driven increase in wages, say due to a successful union campaign or a minimum wage increase, may accrue disproportionately to advantaged workers. If work reorganization either raises an employer's demand for skill or independently raises a job's wage, then it risks exacerbating cumulative advantage.

Why, then, would an employer ever raise wages without simultaneously shifting toward workers that would have earned more anyway? In this paper, we build on an older tradition in organizational research, which theorized the demand-side sources of wage differences within occupations and between organizations (Berg, 1981). This literature identifies two general sources of work organization-linked pay boosts, over and above worker skill and labor market institutions: worker resistance problems, or the translation of potential into actual labor effort; and worker commitment and exit risk (Granovetter, 1981). When employer control over work decreases, employers raise wages to deter worker resistance. When employers upgrade jobs in the direction of company-specific skills, learned by workers on the job, worker exit becomes more costly to the employer. To increase worker commitment, employers raise pay.

These two classic employer dilemmas imply distinct channels through which work organization can affect pay. However, despite interest in demographics and exclusion, key forms of cumulative advantage in the American labor market, this classic organizational and sociological research, along with the related efficiency wage (Shapiro and Stiglitz, 1984; Bowles and Gintis, 1990) and monopsony (Manning, 2003) literatures in economics, have largely stopped at theorizing these channels as sources of high-paying jobs for frontline workers. That is to say, they do not study how wages and worker selection shift as a function of changes in work organization. This leaves uncertain how worker resistance and commitment

affect cumulative disadvantage, rather than pay levels in isolation.

We argue that commitment and consent sources of pay premiums shift the cumulative advantage implications of work organization changes. Specifically, increasing worker autonomy induces no automatic shift towards hiring more advantaged workers: higher-paid alternative job prospects for those workers would erase the premium paid to maintain a high effort. For example, a high-paying, high-motivation job prospect for a high school graduate may be the temporary fallback plan for a college graduate. Similarly, protecting training investments will not push employers toward more advantaged workers: workers with lower-paid alternative job prospects will typically be easier to retain, mapping onto the commitment dilemma. Autonomy and demand for on-the-job skills will thus tend not to drive selection toward higher-skill workers, especially when compared to demand for skills that can only be learned through formal education. Increasing autonomy and on-the-job learning can therefore interrupt the cumulative advantage that otherwise emerges from raising wages.

To test these ideas, we study work reorganization alongside changing worker earnings and worker selection. This requires observing both the shifting task content of jobs and also the changing composition of workers hired into those jobs. To do this, we conduct the first ever merge of online job posts to linked employer-employee earnings data. By matching new hires to the job posts they were hired under, we compare workers hired into the same employer and the same occupation, but under more complex or higher autonomy task allocations. This approach does not give random assignment of tasks to jobs. But, it does allow tight comparisons within jobs, controlling for time-varying labor market contexts, that shift the organization of work. Unlike between-occupation comparisons, or between-company comparisons, we hold constant macro-level features of position and organization. We also focus on the relevant context for the older structural sociology of work, which prior research proxied imperfectly at the industry level: actual jobs.

This data infrastructure lets us directly study how workers' earnings vary with changing work organization within jobs: a key goal of prior organizational research on how reorganizing

work can support higher pay. But, our data also allow us to assess whether pay-increasing work organization changes exacerbate or offset cumulative advantage. Employers can hire different types of workers under different job posts, and we study how worker selection changes as the organization of work in the job shifts. Finally, we follow newly hired workers for 4 years after their initial hire, tracking them across jobs and companies, to assess which types of work organization durably interrupt cumulative advantage.

Together, these analyses contribute to organizational research on labor market stratification, by linking tasks and work organization not only to pay but also to the varying backgrounds of workers hired into upgraded jobs. We bring the classic demand-side perspective in organizational theory into a framework that considers both inequality and exclusion. Distinguishing types of work reorganization reveals different patterns of cumulative advantage and opportunity beneath similar earnings increases.

2 Work Organization and Cumulative Advantage

Concerns about economic opportunity often boil down to a relatively simple question: are workers with greater training, experience, and earning potential, capturing an ever-growing share of economic rewards? Focused on scientific careers, Merton (1988) introduced the concept of cumulative advantage to describe how training, position, and resources help relatively more advantaged scientists accumulate rewards over time. A large subsequent literature develops this concept to explain how early wins can accelerate products' and companies' climb to the top (e.g., Berg, 2022). Building on structural theories of worker mobility (Baron and Bielby, 1980; DiPrete and Eirich, 2006), we study cumulative advantage in the labor market. This work explains mechanisms such as the disproportionately negative impact of rejections on women applicants (Brands and Fernandez-Mateo, 2017), compensating penalties for upwardly mobile but previously disadvantaged workers (Pongeluppe, 2024), or opportunities for training revealed only to insiders (Beane, 2019). In an era of rapid reorganization and technological change, we focus on the organization of work, the assignment of tasks to jobs.

Organizational research has long emphasized that work organization is not merely determined by immutable, labor market-wide technological imperatives. Instead, it is a function of managerial strategies of control (Noble, 1978); decentralized job crafting and assembling (Wrzesniewski and Dutton, 2001; Cohen, 2013); shifting borders of professional jurisdictions (Chown, 2020); interdependencies in job structure (Hasan et al., 2015); managerial value commitments (Ton, 2014) and policy decisions (Fernandez, 2001; Autor et al., 2002); and mental models of occupations (Barley, 1996); and demographics and tasks (Feldberg, 2022; Martin-Caughey, 2021). Indeed, task allocation practices vary across and within organizations for reasons ranging from broad commitments to high-performance work systems (Handel and Levine, 2004) or structured management practices (Bender et al., 2018), to negotiated, dynamic allocations of tasks that vary even within similar jobs in the same organization (Anthony, 2021). Even conditional on some frontier production technology, companies can vary substantially in the details of their work organization (Beckman and Burton, 2008).

This apparent variation in task allocation across companies has driven interest in whether distinct configurations of tasks might provide higher-paying jobs to disadvantaged workers. If the demand side of the labor market is not fixed by exogenous, evenly diffused technology, then the supply side of workers' human capital characteristics does not fully determine their outcomes. Work reorganization in directions that support higher paying frontline jobs could interrupt cumulative disadvantage and provide increased earnings for non-college, low-wage and otherwise marginalized workers.

Consistent with this possibility, a growing literature spanning management, economics and sociology, studies how the organization of work affects pay and inequality. Examples abound. The implementation of high performance work practices, e.g., teamwork and job rotation, is associated with increased pay (Osterman, 2006). Structured management practices are associated with higher productivity and higher pay (Bender et al., 2018; Bloom et al., 2021). Multi-tasking and devolution of management functions increases pay for skilled work-

ers (Caroli and Van Reenen, 2001). Jobs that allow some protected turf are associated with higher pay, relative to those that are redundant with co-workers' jobs (Wilmers, 2020). Together, these findings suggest that some types of work organization could provide higher paying jobs for frontline workers.

Yet, notwithstanding this cross-disciplinary evidence, the possible link between work organization and higher-paying jobs faces a crucial challenge. Even when a task change appears to raise pay, it may do so particularly for higher-skilled workers (Caroli and Van Reenen, 2001). In one case study, job enlargement and intensification actually lowered pay for jobs at the bottom of a plant's pay distribution (Fernandez, 2001). Moreover, upgraded jobs can change the types of workers selecting into a given job, such that previously accessible jobs are filled by more-credentialed workers (Bender et al., 2018; Modestino et al., 2019). Perhaps a retail store assigns some inventory and merchandising responsibilities to its entry-level clerk position and accordingly raises its starting wage. That job might now attract and employ workers who have some retail experience or have completed their high school degree, rather than workers with less than a high school degree. In this case, redesigning the job to be more complex or higher autonomy could raise pay for a given *job*, but actually exacerbate cumulative disadvantage by reducing opportunities for the lowest-wage *workers*.

Addressing this question of cumulative advantage requires clarifying the specific mechanisms that link task changes to higher pay. Some studies attribute the positive wage effects of alternative work organization to upskilling or training (Lynch, 1992). Others point broadly to increased bargaining power associated with higher worker discretion (Osterman, 2006; Wilmers, 2020). Without distinguishing specific forms of work organization, it is difficult to determine how work reorganization will affect cumulative advantage. Indeed, the extant research has typically started from assessing some specific management trend—like high-involvement work, lean production or job enlargement—rather than theorizing the fundamental ways that different work organization shapes pay. What features of work tasks motivate employers to raise wages without shifting toward more expensive workers?

3 Varieties of Job Upgrading

Many cases of employer job upgrading amplify cumulative advantage. Employers typically hire the highest-ability worker willing to accept the job at a given wage. If an employer unilaterally and idiosyncratically increases the wage for a reorganized job, the employer can allocate that job to a relatively more advantaged worker compared to those whom the job attracted prior to the wage increase.

But does all work reorganization follow this positive selection approach? In the following, we distinguish three broad types of work organization changes: those that upgrade jobs by increasing demand for technical skills; those that increase workers' autonomy and discretion; and those that increase employer-specific and on-the-job learning. We argue that the latter two types of work reorganization have the potential to interrupt cumulative advantage, by raising wages for reasons independent of pre-existing worker advantages.

First, consider investment in new technology that automates routine tasks but increases demand for non-routine, technical-cognitive tasks. For example, when programming tasks on lathes are shifted from engineers to machinists, machinists take on more complex, technical tasks (Kelley, 1990). Because formal knowledge is critical for much complex task performance, these skills are typically learned off the job through formal schooling and explicit training. These tasks command a wage premium largely because they typically require formal education (Mincer, 1974) and performing them can signal skill to other employers (Galperin et al., 2020). But shifting toward these technical tasks, in which off-the-job training is typically necessary, is likely to require selecting for higher-ability workers. In one case study, new machinery in a plant retooling was followed by increased hiring of highly paid workers (Fernandez, 2001) and in another, digitization of check imaging at a bank increased demand for college graduates (**?**).

This type of work reorganization also extends to job upgrading beyond technological change narrowly construed: as noted above, prior research finds that many forms of job enlargement (Herzberg, 1968), devolution (Caroli and Van Reenen, 2001), and worker in-

volvement (Osterman and Shulman, 2011) deliver jobs with higher wages and higher productivity, but deliver these jobs specifically to higher-skilled workers. Because employers positively select workers up to the highest offered wage, many forms of job upgrading, while raising wages within a job, may actually reduce demand for disadvantaged workers. A job that was previously accessible to a worker at the bottom of the labor market ends up going to a worker who already has a leg up.

Are there any types of work reorganization that would raise wages without exacerbating cumulative advantage? Why would an employer ever rejigger work organization in a way that requires higher wages without substituting more expensive workers? To develop an answer, we return to an older line of organizational research that emphasized employer-level demand-side theories of pay (Berg, 1981) in contrast to human capital (Becker, 1962) and status attainment (Davis and Moore, 1945) accounts of earnings and inequality. Alongside employer choices about the complexity of jobs and their worker ability requirements, this line of research also considered two additional sources of pay premiums related to work organization: worker resistance and worker commitment.

First, employers face the problem of eliciting consent from workers with interests often opposed to the employer's goals (Granovetter, 1981). Unlike renting office space or purchasing a computer, an employer hiring a worker must contend with the workers' interests. Research in the sociology of work identifies one response to this problem: employers adopt a detailed division of labor, coupled with heavy use of supervision or automated surveillance, seeking to eliminate worker discretion. Different flavors of autonomy-reduction may rely more on direct supervision, bureaucratic rules, or on technical or algorithmic control (Edwards, 1979; Marglin, 2015). Classic and contemporary organizational research highlights an even broader set of reasons that employers might reduce worker autonomy: Shifting frontline workers towards more easily supervisable or rule-bound tasks, for example, has been linked to fewer mistakes and the acceleration of information sharing (Katz, 1965; Scott, 2004; Parker et al., 2017; Foss and Klein, 2023). Some of these performance improvements hinge on lower

autonomy reducing variation in work process through the company, but others may indeed flow through a reduction in worker ability to resist managerial demands.

An alternative approach lies in fostering workers' responsible autonomy. This harnesses worker ingenuity and discretion to achieve locally optimal responses to work challenges. US manufacturers embracing lean production, for example, empower production workers to identify and even resolve production problems as they arise, as in the use of the Andon cord in the Toyota Production System (Womack et al., 1991; Vidal, 2022). Likewise, some call centers shift toward increasing autonomy and discretion by reducing the use of scripts for customer service agents, who instead are relied on to use discretion in responding to idiosyncratic customer concerns (Rafaeli et al., 2008). More broadly, assigning workers tasks that increase their responsibility over their work, may benefit *employers*, by accelerating decision-making (Balachandran and Eklund, 2024), increasing satisfaction and retention (Cameron, 2024), complementing other engagement practices (Young-Hyman, 2017), reducing resistance to employer initiatives (Kellogg et al., 2020), and avoiding costly worker behavior to evade monitoring (Bernstein, 2012). Across these varied case study contexts, the classic ambiguities of organic organizations (Burns and Stalker, 1961) and normative management (Barley and Kunda, 1992) emerge for work organization in which employers do not use standard monitoring, direct control or bureaucratic rules. As Sorenson and Kalleberg (1981) put it, "The employee's control over information about the job thus represents a resource that may be used to further the employee's interest" (p.64).

Regardless, embracing worker empowerment requires aligning workers' incentives with those of the employer. Paying wages above what a worker would make elsewhere provides those incentives (Shapiro and Stiglitz, 1984). Specifically, if a worker is paid above the wage at their next best alternative job, and they are caught using their autonomy counterproductively (from the employer's perspective), they could lose their high-paying job. In this way, an efficiency wage directs worker discretion toward effective task completion. Note that here, a higher wage is not paid to attract higher skill workers. Indeed, consider the case of an em-

ployer increasing job autonomy, while also hiring a more advantaged worker. The employer would have to further increase the wage, relative to hiring a worker of similar skill level in the less autonomous version of a job. Otherwise, the newly hired, more advantaged worker would not receive the incentive-aligning extra pay necessitated by autonomous job design. Higher wages emerge here to reduce the productivity costs arising from worker resistance. An autonomy pay boost is effective only when set above a worker's alternative employment prospects, regardless of the value of those prospects.

Second, employers face the problem of worker commitment. Outside of the gig economy or seasonal farm labor, the employment relationship is a long way from a spot transaction contract. Workers are hired to do a job, often on an indefinite basis. One reason for this lies in the distinctive features of any workplace: workers learn on the job, and cannot be immediately replaced by a new hire. However, employers have no gaurantee that workers will stay for the long-term. Turnover is costly (Kuhn and Yu, 2021). As such, ensuring long-term worker commitment becomes a goal for many employers.

Indeed, it becomes a goal particularly for employers who organize work to require substantial on-the-job learning (Althauser and Kalleberg, 1981, 126). When employers reorganize work to require more complex, but organizationally idiosyncratic, tasks, they may increase productivity and value-add. For example, when hospitals add care coordination to medical administrative assistants' work (Kottek et al., 2021) or when manufacturers give maintenance and process improvement responsibilities to production line workers (Helper and Kuan, 2017; Adler et al., 1999). However, this type of work organization increases employers' reliance on workers who stick around long enough to learn the job and move up a tenure-productivity curve (Doeringer and Piore, 1971). With employer-specific training, employers invest "in the employee, and the employee gains control over access to the job in proportion to the size of the investment" (Sorenson and Kalleberg, 1981). Employers must thereby increase worker commitment using the same strategy that minimizes worker resistance: raising wages above what a competitor would pay that worker. High-wage employers can harbor more complex

production processes in part because they can retain workers more reliably.

Here, though, as with autonomy-resistance, the extra pay to reduce turnover must be relative to a worker of a given opportunity level. The employer thereby faces no new opportunity to hire a more advantaged worker as she embraces on-the-job learning intensive job design. Shifting toward higher skill workers will require additional extra pay to maintain the level of commitment elicited from a wage premium designed for a less advantaged worker.

The key underlying mechanism in these resistance and commitment sources of higher pay is that a wage is set at a premium relative to a worker's alternative employment opportunities. If an employer hires a worker with more credentials or experience into a newly reorganized role, the wage will have to rise additionally. If the next best option for an entry-level worker amounts to $8, perhaps a wage that minimizes worker resistance would be $9. But if an employer posted the job at $9 and instead hired a candidate with valuable experience that raised her best alternative offer to $8.50, then the no-resistance wage would become $9.50 (assuming that the costs of effort for the entry-level and experienced worker are equivalent). As a result, when an employer raises pay to reduce worker resistance or increase worker commitment, they have no new opportunity to select more qualified workers relative to those hired prior to the work reorganization and pay increase. This process interrupts the standard positive selection induced by versions of work reorganization that increase task complexity and skill demand. By extension, this mechanism, of paying a premium above a worker's alternatives, interrupts cumulative advantage in the labor market. Rather than disproportionately allocating newly higher paying jobs to more advantaged workers, this mechanism assigns them to workers similar to those who did the job prior to its upgrading.

4 Studying Work Organization

This theory distinguishing dimensions of work organization by their effects on labor market cumulative advantage has not been previously tested. This is due in part to a limited research infrastructure linking tasks and pay for a broad worker sample, and in part to challenges in

abstracting broader mechanisms from the hugely dimensional space of all work tasks across the economy. Before turning to the details of our research design, we discuss why the three main approaches to studying tasks—occupational task coding, establishment-level surveys, and employer- or industry-specific administrative data—are inadequate to test these ideas.

The standard approach to studying tasks uses Dictionary of Occupation Titles or O*NET data to categorize occupations according to their task content (Autor and Handel, 2013; Liu and Grusky, 2013). This work cannot measure the substantial task heterogeneity within occupations emphasized in recent work (Martin-Caughey, 2021). It therefore misses the key margin of employer decision making about job design: a health center would rarely choose whether to hire a doctor or a community health worker, but the center could have substantial leeway in determining which tasks should define the community health worker's responsibilities (Kellogg, 2014). Likewise, a manufacturing facility will necessarily employ both engineers and assemblers, but might choose exactly how much engagement assemblers, rather than engineers, have in solving unexpected production problems (Bechky, 2003). While this work therefore makes sense when considering the shifting economy overall, the most actionable test of our theory from a manager's perspective would track how work organization varies within occupations.

The standard way of capturing this within-occupation variation has been to field surveys of managers, asking about establishment-specific work organization. Researchers have used this method to study work organization systems like high-involvement work practices (Osterman, 2006), joint-decision making (Black and Lynch, 2001), structured management practices (Bloom and Van Reenen, 2007), job design and enhancement initiatives (Carpini et al., 2017), and cross-functional and team-based work organization (Kalev, 2009). These surveys have been revealing, but they assume that work organization is constant within establishments (or varies only within broad categories of jobs, like white collar vs. blue collar). This prevents them from capturing details of task composition, which are inherently job- and occupation-specific. This approach also struggles to study worker sorting: a

cross-sectional establishment-level survey cannot rigorously distinguish apparent pay effects of work organization from changing worker selection.

Progress has recently been made on these issues by organizational scholars using employer- and industry-level administrative task data (Fernandez, 2001; Wilmers, 2020; Chown, 2020; Ranganathan, 2023). These projects can directly observe within-organization variation in tasks, often with detailed information about job titles and organizational structure. However, because they rely on employer- or industry-specific data, they cannot track workers as they move across employers or in and out of a focus industry. This means that studying selection and the dynamics of cumulative advantage across a workers' career is not possible.

In contrast to these approaches, we implement a novel research design that allows us to observe detailed information about task allocation, from job post, and worker earnings and characteristics, from labor market-wide administrative records. This design brings the granularity missing from occupation-level analyses; addresses the job specificity problems in establishment surveys; and by tracking workers across jobs, allows detailed analysis of selection. To match new hires to the job posts under which they were hired, we merge a panel of job descriptions to linked employer-employee earnings data. This links high-quality, worker-side selection data to a specific task set that the worker is hired to do. It also allows a tight comparison between workers hired into the same employer and the same occupation, but with different task allocations.

In the following, we describe the substantial data infrastructure work necessary to develop this research design. But, first we note the key drawback of our approach: we do not have direct measures of workers' tasks, but only what is advertised by employers. If employers omit important responsibilities or distort the content of work, that introduces noise into our measure of work organization. We discuss this issue further below, but we see this as a limitation in our project that is counterbalanced by gains in granularity, tight comparisons, and analysis of shifting worker selection.

5 Data and Measures

We study how work organization shapes cumulative advantage by comparing the earnings and selection of workers hired into similar positions with different task allocations. To do this, we link U.S. administrative earnings records of workers starting jobs, to the tasks that appear in online job posts at the same time and in the same employer, occupation, and commuting zone. We then measure the task complexity and autonomy of each job post, sourced from Burning Glass (BG), through an original survey asking respondents to rate job tasks common in their occupational group. We follow individual worker earnings outcomes by merging these survey-coded BG data with longitudinally-linked worker and employer data from the U.S. Census Longitudinal Household-Employer Dynamics (LEHD) database, using occupation and education information from the American Community Survey (ACS). This is the first study to link the rich task information from job posts to administrative earnings records. For a wide variety of employers, workers, and occupations, we track how changes in tasks within employer-by-occupation positions affect employer hiring and worker earnings.

5.1 Burning Glass Job Posts

To build their job post data, BG draws from over 40,000 online job boards and employer websites, removes duplicate posts, and parses the resulting data into a standardized database (Deming and Kahn, 2018; Hershbein and Kahn, 2018; Wilmers and Zhang, 2022). Although online job posts do not capture all openings, a recent validation by Acemoglu et al. (2022) finds that the BG data closely align with trends in overall U.S. job vacancies. Earlier work suggests that 60-70% of U.S. private sector job openings in recent years have been posted online (Carnevale et al., 2014), and this rate has likely increased over the intervening decade.

To record the task requirements described by each job post, we use BG's task coding of job post text describing job tasks and responsibilities. BG identifies 16,128 unique task requirements ranging from technical activities such as *coding in Python* to physical work such as *lifting heavy objects* to generic cognitive requirements like *analytical thinking*. This

BG task typology is used consistently throughout our 2010-2021 study period.

To identify these task requirements in job post texts, BG uses natural language processing classifiers comprising thousands of rules and keywords, including numerous neighborhood and negation rules. These rules allow the algorithm to distinguish homonyms such as "kitchen chef" and the "chef" software tool. Beyond homonyms, BG's algorithm (guided by hand review) recodes similar *task concepts* appearing across the job post texts into 16,128 tasks. For example, synonyms such as "customer engagement" and "customer relations," are consolidated into the same BG task. The distribution of appearances of unique tasks has a long right tail, for example, as a few tasks that are general to many occupations will appear much more frequently than a more specialized tasks.

After necessary exclusions, we obtain a sample of 150 million job posts each associated with 8 tasks on average, totaling 1.3 billion appearances of the 16,128 BG tasks. The largest sample exclusion relates to employer name, appearing in 73% of the 246 million posts with some text-based skill information (for a discussion of this, see Hershbein and Kahn, 2018). Ninety-five percent of the remaining posts include occupation and county. Occupation is coded by BG from job title and county is coded from the employer workplace address (we describe how our study would be different if we were able to match on job title rather than occupation, in the Discussion section 8). Eighty-four percent of the remaining posts have skill information based on job descriptions.

Our resulting job post sample covers 99% of SOC occupations across all U.S. commuting zones over 12 years. Table A.1, discussed below, shows that managerial and technical roles are overrepresented in the BG data (Acemoglu et al., 2022), but that frontline and supervisor job posts are substantially represented: 75 of the 150 million job posts that we use are for non-managerial, non-professional occupations, with an equivalent national employment figure of 93 million. We discuss below how we weight our regressions to adjust our sample to U.S. national occupational and demographic proportions, and we also run our regressions keeping only non-managerial, non-professional workers. Below, we also compare the occupation and

industrial distribution of our merged data and the BG posts with the ACS and the Current Population Survey - Annual Social and Economic Supplement (ASEC) two leading nationally representative social surveys.

5.2 Survey to Classify Jobs Tasks on Dimensions of Work Organization

The BG processed job posts thus include rich information about work tasks. However, the thousands of discrete tasks appearing in the job posts do not naturally correspond to the general conceptual framework discussed above and which we seek to test. As such, we field two novel surveys asking respondents to categorize job tasks into types of tasks. We then take the average of each job's task scores to measure the complexity and autonomy of that job, the main predictors needed to assess our theory of work organization and pay. We post our survey results and all code for this project at [TK post-review]. Appendix A provides technical details on the surveys we ran.

In our surveys, respondents categorize a subset of BG task requirements on one of 2 dimensions characterized by 3 response options each. In the first version of the survey, respondents are asked to map task complexity. Here, we sought to distinguish types of job upgrading that rely more on technical skills, acquired through formal schooling and marking advantaged workers, from types of job upgrading that reflect skills learned informally on the job. Specifically, respondents report whether a task is (i) complex and likely learned only off the job, (ii) complex and possible to be learned on the job, or (iii) non-complex or simple. Appendix A provides the full question text used to prompt these concepts.

In the second version of the survey, we map the degree of control or autonomy implied by a task. Here we started with a simple contrast between high autonomy or discretion tasks and tasks that tend to be easily supervised and tightly controlled. However, looking through task lists, and examining respondent feedback in an initial pilot, we noted another mode of constraint, via relationships and interactions with others. This idea corresponds to customer (Sherman, 2011) and concertive (Barker, 1993) control studied widely in the sociology of

work. For our final categorization, we asked respondents whether a task incorporates (i) high discretion and autonomy, (ii) constraint through interaction with coworkers, clients, or customers, or (iii) high rules and/or supervision. This allows us to distinguish high discretion jobs from those that are constrained either by rules and tight supervision or by customer or coworker interaction.

In total, we collect 120,000 task-by-dimension response values: 30 responses for each of 2,000 tasks on both dimensions. We focus on only the most common 2,000 tasks that make up 93% of all task appearances in the BG data. Appendix A provides a technical overview and an example of the complexity and autonomy surveys. In addition to standard attention checks and participant screening, we assign respondents only to tasks that were most frequently in their main occupation; reduce noise by collecting 30 responses per task and per dimension; and train respondents interactively, providing instructions and feedback for categorizing tasks along the complexity and autonomy dimensions.

To transform the resulting task-level survey codings into job-post-level task allocation scores, we take the average of the task-level scores for each job post. Consider a post associated with tasks A through E. If 90% (27/30) of respondents chose high autonomy for task A, and B, C, D, and E were analogously given 0%, 10%, 10%, and 40% discretion scores, respectively, through our survey. Our discretion measure for that job would be $(90+0+10+10+40)/(5) = .33$. In this way, the "discretion," "relational control," and "supervisable" scores for any job post will sum to 1 by design, as will that post's "off-the-job complexity," "on-the-job complexity," and "simple" scores. Therefore, we use 4 variables as predictors: discretion and relational control, which are compared to the supervisable reference category, and off-the-job complexity and on-the-job complexity, which are compared to the simple task reference category. Using the results of our survey covering the top 2,000 BG tasks, we code 93% of task appearances and job-level measures for 99% of jobs, giving us complexity and autonomy measures for over 160 million BG job posts.

As an example of our task coding, we compare two line cook posts from 2014 and 2021,

respectively, both posted in Eastern Pennsylvania by Sodexo, a large food service company. The 2014 post contains the tasks *cooking*, *food preparation*, *meal preparation*, and *physical abilities*, while the 2021 post has all four of these tasks as well as *building effective relationships*, *work area maintenance*, and *Hazard Analysis Critical Control Point (HACCP)*. These three tasks signal an increase in relational control and complex tasks. In terms of our scores, the job changes from 0.27 to 0.38 on-the-job complexity, 0.08 to 0.23 off-the-job complexity, and 0.1 to 0.21 bound by relationships. This example highlights how our scores are based on averages across skills in the job post. The pure addition of some skills can reduce scores as the prior listed skills theoretically receive less emphasis. This example also highlights how we allow tasks to have intermediate values on our work allocation dimensions: *food preparation* is 0.37 on-the-job complexity, 0.05 off-the-job complexity, 0.19 autonomy, and 0.04 relational control, and therefore 0.58 simple and 0.77 easily supervisable, signaling varied answers among respondents for this task.

To further illustrate the task codings, Figure 1 plots a random sample of BG tasks based on their average scores on the complexity and autonomy dimensions. In the upper panel, representing complexity, the origin contains simple, low-complexity tasks; the y-axis, labeled on-the-job, represents greater complexity tasks that can be learned on the job; and the x-axis, labeled off-the-job, represents greater complexity tasks that respondents believed required formal education. Complex off-the-job tasks appear to mostly be associated with skills taught in higher education. For example, "maintaining supplies" has a relatively low on-the-job complexity score and a minimal off-the-job complexity score, "freight flow" analysis is complex but can be learned on the job, and all respondents agreed that "geometric dimensioning," an engineering design system, would be learned off the job. In general, we see most tasks sitting between on-the-job and off-the-job complexity. The overall wedge structure of the results plotted on these axes suggests a general linear tendency from simple tasks, through on-the-job complexity, and to off-the-job complexity: it was rare that survey respondents were split between whether a task was simple or complex off-the-job.

[Figure 1 about here.]

The bottom panel of Figure 1 plots BG tasks based on their average scores on the autonomy dimension. The origin represents rule-bound or easily supervised tasks that are likely constrained by detailed rules; the y-axis, labeled relational, represents tasks in which worker autonomy is constrained by interaction with other people inside or outside the organization; and the x-axis, labeled discretion, reflects tasks that require high autonomy and independent decision-making. For example, "X-ray" and the "Omniture" business software score as easily supervised tasks; "client base retention" and "customer referrals" are highly relational; "creative problem solving" is maxed out on discretion; and tasks like "clinical development" and "performance analysis" rank relatively low on relational control but intermediate on autonomy. Most tasks sit in-between the origin and full discretion, indicating relatively fewer relational constraint survey responses. Overall, these tests suggest that our novel survey-based encoding captures the core theoretical dimensions we introduced above.

5.3 LEHD-ACS: Bringing Occupation to Administrative Earnings Data

To complement the survey-coded BG task measures, we draw earnings and employment information from the restricted-access U.S. Census Longitudinal Employer-Household Dynamics (LEHD) database. The Census Bureau maintains the LEHD by integrating quarterly reports from employers to U.S. state unemployment insurance administrations (Graham et al., 2022). These data are combined with Internal Revenue Service and Social Security Administration records. The resulting longitudinally-linked data describe the quarterly earnings for 97% of U.S. private sector workers from 1996Q1 through 2022Q1. By "earnings," we refer to compensation broadly, rather than base pay rates narrowly construed: the LEHD covers salary, cash, commissions, bonuses, tips, awards, severance, vacation pay, standby pay, reimbursements, back pay, sick pay, and payments made under a deferred compensation, with some minor variation across states. We accessed these data through the Census Research Data Center program that grants non-Census researchers FBI Special Sworn Status to use

confidential U.S. administrative data.

In Appendix B we describe a series of standard restrictions to the data: academic researchers receive a random sample of half of US states; we drop earnings quarters at less than half time minimum wage; and we require jobs to span at least 3 quarters, so that the second employment quarter is guaranteed to be a full quarter of earnings.

Note that unlike employer-specific data, the LEHD follow individual workers, even as they change employers over time. We use the LEHD data to construct two worker earnings outcome measures, plus four work history control variables. Our main outcome, *second quarter earnings*, which we log, requires no additional processing and is the first full employment quarter for a worker at a given employer-occupation job. To test effect persistence, we also predict worker earnings 4 years after job start. We measure this as the average of observed logged earnings 16 and 17 quarters after start. Observing this long-term outcome mechanically excludes workers who start after the fourth quarter of 2017, plus some workers who are not in the labor force in our covered states at the measurement time, though we compare the main and long-term samples below. Note that here we do include workers who exit the firm or job they are initially hired into, which allows us to assess a true long-term trajectory, unbiased by selection out of, for example, a single organization or industry.

Second, we use the LEHD earnings data to construct four work history human capital control variables, discussed in the Models section below. These are a crucial set of measures for assessing cumulative advantage, as they allow us to characterize newly hired workers by their earnings level and trajectory in previous jobs.

Occupation is a critical aspect of our merging strategy, allowing comparisons of task allocation and pay within job positions. As the LEHD data contain no occupation information, we merge in the American Community Survey (ACS), a Census Bureau survey of over 5 million U.S. residents per year. The ACS also provides demographic measures including age, demographics, and education (while the LEHD contains education information, it groups all workers with a BA, whereas we want to differentiate those with graduate degrees). We link

to the LEHD through a Census crosswalk based on Social Security Numbers.[1] Specifically, we only include LEHD job starters sampled by the ACS within an 8-quarter window before or after the job start quarter.

The the ACS merge allows us to attach occupation to the LEHD earnings records, and reduces the sample size of job starts by 99%, randomly except for oversampling cohabitating adults, which we adjust for using Census-provided regression weights. The ACS match, based on worker and timing, cuts our LEHD job start sample from hundreds of millions to 6,624,000 job starts.[2] Note that this merging step imposes very little selection pressure on the coverage of job starts in our sample: The ACS is a high-quality random sample of households, so the subset of job starters who match to the ACS in a given time window is essentially a random set of job starters after applying the ACS person weights.

5.4 Merging the BG to the LEHD-ACS Data

We merge BG job posts to LEHD-ACS job starts based on employer, occupation, commuting zone, and time frame. As our project is the first to merge these datasets, we summarize the two steps of this merge process here and discuss it in more detail in Appendix C.

First, we require a match on employer name with evidence that the employers on each side operated in at least one shared commuting zone during the study period. Appendix C introduces the key challenges in matching BG to Census data, for example different audiences (job seeker versus governmental) or conceptions of aggregation (concerning franchise or business units). Appendix C then describes our work to overcome these challenges through data cleaning, fuzzy matching, and cleaning the employer-occupation-(commuting zone) matched units. We aim to be strict in name matching to reduce attenuation bias in our estimates due to noise from false matches. Even with strict name and location match requirements,

[1] We also integrate responses from the 2008-2020 Current Population Survey Annual Social and Economic Supplements (CPS). We recode all CPS variables to match their ACS equivalents. The CPS sample size is less than 2% of the ACS sample.

[2] All sample size figures are rounded in this document according to Census rules where applicable. For comparison, the Bureau of Labor Statistics JOLTS survey would roughly estimate 300 million hires overall in our 24 covered states per year during the eleven year study period.

we match 57% of BG job posts on employer (86/150 million), moving us from 6.6 million initial ACS-LEHD job starts to 3.8 million job starts (also 57%).

Second, we require a match on occupation, commuting zone, and time frame. For occupation, we translate each ACS OCC code into its corresponding 5-digit SOC code (used by BG), with a crosswalk from the Integrated Public Use Microdata Series website, standardized over time (Ruggles et al., 2020). For commuting zone, we use the county of the BG job post workplace county and the ACS reported workplace county, using the LEHD Individual Characteristics Title 26 files when ACS workplace county is rarely missing (Vilhuber et al., 2014). Commuting zones were defined by the Census based on residence-to-work commuting flows from the 2000 Decennial Census. For time frame, the majority of our LEHD-ACS-BG-merged job starts use job posts in the quarter immediately before the job start quarter, but we allow for a match up to 3 quarters before job start or in the same quarter as the job start.

More specifically on time frame, we first average our work organization scores to the employer-(commuting zone)-occupation-quarter level. When the quarter before does not exist we prefer matching the job post information from -2, 0, -3, and then -4 quarters before the job start, allowing for interviewing and onboarding time (we present robustness results preferring posts in the same quarter as the job start). We choose this fairly tight time frame because, while a post 2 years after a job start might be a good measure of tasks undertaken by a starting worker, these temporal imputations will obscure within-occupation-employer comparisons over time, critical to our goal of studying *changes* in work organization, for example for retail workers at employer X over time.

After adding these requirements, dropping workers without work history, and dropping employer-occupation units that only appear once in our BG-matched job start data (per Census results disclosure rules to achieve 1 sample across our regressions), we have 611,000 employer-occupation-(commuting zone)-time-matched job starts in our main sample, comprising 570,000 unique workers and 23,500 employers. We note that our additional requirement of start-post matches in the same occupation and commuting zone (CZ), and within

a short time frame, increases the validity of our study by helping to weed out false name matches, thereby reducing attenuation bias.

To be clear, this approach assigns worker job starts to relevant posts without requiring an exact and exclusive match between starts and posts: If employer X in Greater Boston posts two different ads for retail supervisors in the same quarter, we will average the work organization scores on those two ads before assigning those scores to one or more starting retail supervisors. Conversely, multiple starters can be assigned the scores from the same post, so long as they start in the same occupation, employer, and commuting zone in the acceptable time frame.

5.5 Generalizability, Regression Weighting, and Key Sample Features

The resulting dataset matches surveyed workers to their full set of administrative earnings records and to the employer-occupation-CZ job posting active around the quarter they were hired. The underlying data for these merged records is the population of worker earnings records, a random sample of those workers and the near-population of online job postings. However, the online-only nature of the job postings biases them toward managerial-professional jobs, as noted above, and the merge process itself can introduce additional sources of bias. As such, in this section, we explain how and why we apply weights to the matched data and introduce potential implications of persistent bias in the sample, before discuss key descriptive statistics.

We use regression weighting to address the generalizability concern arising from an over-representation of managerial and technical workers in our merged BG-LEHD data. Specifically, we aim to ensure that our estimates reflect U.S. national demographic and occupational proportions. We also confirm our results to keeping only non-managerial, non-technical workers (see Table A.4).

Ideally, our study would randomly sample workers across all jobs, observing pay and our work organization predictors. This would make it generalizable to the U.S. workforce at a

given time. However, because we are limited to observing work organization via job postings, this strictly worker-based sampling approach is not possible. Because posts for managerial and professional workers are overrepresented in the BG data (see Appendix D), our matched sample also overrepresents these workers. If the relationships between our work organization predictors and pay are substantially different for higher skilled workers than for workers overall, we would generate a non-representative estimate viz a viz all U.S. workers, and, in extreme cases, could falsely reject a true null average relationship. The clearest concern here is that the types of high discretion or complex tasks in managerial and professional occupations may be associated with greater percentage pay increases than the types of high discretion or complex tasks in the remaining 60% of the U.S. workforce.

In addition to this main source of sample bias, there are several additional reasons why our merged BG-LEHD data might diverge from an ideal random sample drawn from the U.S. workforce: (i) focusing on job starts overrepresents the types of workers who change jobs more frequently; (ii) requiring work history focuses on continuing workers; (iii) our 24 LEHD states may be different from the U.S. overall; (iv) patterns in Census and BG employer names may lead to the overrepresentation of some workers in hard-to-predict ways; and, (v) our research design of comparing the work organization and pay relationships within employer-occupation positions over time, given a 1% random job start sample based on the ACS merge, leads to the overrepresentation of the kinds of workers starting jobs at larger employers.

We address this divergence from random sampling empirically, by giving more weight in our regressions to job starts for the types of workers that are underrepresented in our merged BG-LEHD data, relative to U.S. national demographic and occupational proportions. Specifically, we follow the prior literature (Lemieux, 2006) to fit a logistic regression on all ACS responses of employed workers during the study period:

$$in_i = s_i + a_i + a_i^2 + r_i + e_i + o_i \ ,$$

where in_i captures whether worker i appears in our merged BG-LEHD data; s_i is binary for female; a_i is age in years at time of job start; r_i captures White, Black, Hispanic, or Asian/other; e_i captures high school or less, some college or an associates degree, a four-year college degree, or a graduate degree; and o_i captures 2-digit SOC group. We then use the inverse of the fitted values from this regression for each worker in our main sample (divided by the number of starts observed for that worker) as a weight in our main regressions, ensuring that our estimates reflect similar proportions of worker types as if the sample had been randomly drawn from the U.S. workforce (in relation to these characteristics). A higher fitted value, representing a greater than random likelihood of being in our merged BG-LEHD data, will result in a lower contribution to the regression estimate.

Appendix D explains that while our main sample before weighting is disproportionately higher-educated and in managerial and professional occupations, our weighted samples mirror U.S. national demographic and occupational proportions. Appendix D also shows that our weighting does little to reduce the overrepresentation of job starts at large employers. Whereas the theoretical concern around the selection and pay implications of discretion and complexity across *occupations* is clearer (and weighting on demographics relates directly to studying changes in worker type selection), the potential bias arising from variation in the relationship between our work organization dimensions and pay is more difficult to interpret.

The overrepresentation of large employers would be concerning if larger versus small employers showed different pay premia and selection patterns based on our discretion and complexity measures, controlling for occupation, work history, local labor market characteristics, sector, and employer size factors discussed in the model section below, and for time-invariant company and job position features. Indeed, large employers would generally be expected to set pay more rigidly, which would make our estimates substantively conservative (i.e., we would expect the estimated relationships between work organization and pay to be lower there than in the kinds of employers that are under-represented in our sample). That said, unmeasured employer (and worker) characteristics that may not balance between

our main sample and a randomly drawn sample, for example monitoring capability, may indeed impose heterogeneity bias in averaging the relationship between our our work organization dimensions and pay towards random sampling, though one must assume fairly high correlations between these unmeasured factors and steps in our sample inclusion process, for this to substantively shape the results. We call for future work explaining or more directly modeling effect heterogeneity by employer type and other contextual factors in discussion section 8.

Appendix E presents descriptive statistics on our key study variables. As referenced above, our original survey suggests a substantial prevalence of jobs high in our two complexity measures and discretion (.42, .48, and .37 of all BG job posts, respectively). Job starts in our unweighted sample have somewhat lower complexity (but not discretion) compared to BG job posts overall, though we expect our sample to be closer to random than BG alone because we are combining it with the randomized ACS survey. Weighting reduces the rate of off-the-job complexity in our sample, aligning with a shift away from higher-educated workers. This does not affect the rates of on-the-job complexity or discretion, suggesting a fairly similar distribution of these job characteristics between the kinds of managerial and professional job starts that we down-weight and the job starts we up-weight. Moving to the main dependent variable, weighting reduces average pay as expected, and reduces the recent upward mobility for workers in our sample. Our dependent variable mean of $9,000, about 15% lower than the national full-time worker average, reflects the inclusion of some part-time workers in our sample. Workers on average are receiving a pay boost of about 2-3% (compared to prior six-year average earnings, and removing 0's and transition quarters), suggesting that not all moves are motivated by pay alone.

5.6 Which Employers Offer Higher Complexity and Autonomy Jobs?

These merged data allow the first ever analysis of the types of companies that disproportionately create more complex and higher autonomy frontline jobs. Before proceeding to our

main analysis of the earnings and selection effects of these forms of job upgrading, we contextualize these job characteristics in the rich organizational characteristics data available through the merge to Census data.

Specifically, we here aggregate the data to the employer level. For this analysis, we match companies rather than specific job starts. So, we require companies to match by employer name and to have at least one occupation posted in the same commuting zone as an ACS worker start. Unlike for our main analysis, we do not impose any time constraint on this match, which increases our sample of companies from 23,500 to 59,000. For the resulting matched companies, we then use all non-managerial/professional job posts, not restricting to those that match to a specific ACS worker start, in order to more accurately characterize the general task organization of the companies' frontline jobs. We then compare companies on administrative financial attributes drawn from the Longitudinal Business Database (LBD): employment, total annual pay, annual revenue, and profit (estimated as revenue minus labor costs). We also compare by employer age, employer type, industry and several LEHD-derived variables (earnings, earnings inequality). Table 1 presents the sample means for key variables in this sample, as well as the difference in average value between companies that are above versus below the median of each dimension of work organization.

[Table 1 about here.]

Higher off-the-job complexity employers are large with growing revenue but shrinking employment. This suggests rising productivity for these companies, which is already relatively high (measured roughly as revenue/worker). These companies also pay relatively high earnings overall, not just for frontline jobs, which would be consistent with the positive selection of workers into these more complex tasks hypothesized above. Employers with high off-the-job complexity for frontline workers are concentrated in the business services and health industries as C-corporations and non-profits. These industries have high technical requirements, echoing the examples of off-the-job complex tasks noted above. These em-

ployers are also relatively newer, consistent with the technical tasks that define off-the-job complexity.

Employers with high on-the-job complexity for frontline workers tend to be even larger and have both declining employment and declining revenue growth. Unlike off-the-job complexity employers, these companies are disproportionately older and in mature industries like manufacturing and retail. The tacit and craft knowledge involved in those industries provides opportunity for relatively complex task allocation to frontline workers. These companies (disproportionately C-corps) have high average pay and the largest revenue and profits per employee but face shrinking revenue and employment. These patterns are suggestive of the long-term decline in internal labor markets and employer-specific skills (Lazonick, 1990).

Finally, employers with higher autonomy jobs are substantially younger and around one third smaller than average. These employers have higher pay and productivity, but are less profitable than average. They are also expanding both employment and revenue and increasing productivity over time. These young, fast-growing companies have a relatively higher labor share of total revenue, with above average payroll and below average revenue. They are also in labor-intensive industries like education and social services and business services. These companies, which create jobs with the autonomy tasks hypothesized above to interrupt cumulative advantage, pay well and are expanding.

Many of these differences across companies are attributable to industry differences. However, in Table A.6, we residualize by 2-digit industry, and overall find similar patterns. Note also that across these forms of upgraded work organization, only C-corporations and nonprofits are overrepresented. Frontline jobs in S-corps and partnerships are disproportionately composed of simple and easily supervised or rule-bound tasks.

In sum, this descriptive analysis characterizes the employer attributes associated with more complex and high autonomy tasks assigned to frontline workers. It shows that a broad range of companies experiment with this approach. But it also reveals key differences between companies that focus on one or another type of work organization. Higher autonomy

tasks assigned to frontline workers are disproportionately present at fast growing, young companies. These descriptive patterns highlight the risks of prior work that studies task effects using between-company variation: different types of job design emerge in companies and industries that are different on a number of clustered dimensions. As such, we focus in our main analysis on within-job variation in work organization that avoids the form of employer type bias uncovered here.

6 Model Specification

We use fixed effect regression models to study the relationship between work organization – job complexity and autonomy – and pay, as well as worker selection. Our empirical goal is to make tight comparisons within employer-by-occupation job positions, controlling for shifting local labor market conditions, for example that may induce pay raises beyond managerial control. The full form of these models is:

$$y_{i,j} = \mathbf{u_j'}\beta + \alpha_{SOC5 \wedge Employer} + \alpha_{Start_Q \wedge CZ \wedge SOC2} + \mathbf{x_{i,j}'}\lambda + \epsilon_{i,j} \quad (1)$$

where $y_{i,j}$ measures second quarter earnings for worker i at job j. Our main predictors of interest, $\mathbf{u_j'}$, represent our 4 work organization scores: on-the-job complexity versus off-the-job complexity, which are compared with the reference category of simple, and discretion versus relational constraints, which are compared with the reference category of easily supervisable. Below we discuss how $\alpha_{SOC5 \wedge Employer}$ represents fixed effects for worker job positions; $\alpha_{Start_Q \wedge CZ \wedge SOC2}$ captures local labor market conditions; and $\mathbf{x_{i,j}'}$ measures a series of human capital controls. The residual term $\epsilon_{i,j}$ reflects pay variance not explained by these predictors. We weight our regressions using our ACS-derived job-start-level factor and we cluster standard errors at the employer level to match the level at which broad work organization strategy may be assigned by managers.

The second term in equation 1, $\alpha_{SOC5 \wedge Employer}$, represents fixed effects for worker job

positions. Including this term restricts our comparison to be within the same occupation at the same employer. These comparisons may be made over time or across workplaces in the same employer. This term controls for unmeasured, time-invariant features of job positions such as a legacy of unionization, a durable organizational ethos dedicated to greater investment in workers, or a specific human resource culture determining job post text, which may create a spurious correlation between our measures of work organization and pay.

The third term in equation 1, $\alpha_{Start_Q \wedge CZ \wedge SOC2}$, captures local labor market conditions. These fixed effects absorb all variation due to time-(commuting zone)-(industrial group) units. We here want to control away, for example, labor market tightening simultaneously raising complexity and/or autonomy as well as pay. Including these controls is also important because our core estimates are in part based on geographic variation in task allocation and pay within the same employer: we do not want unmeasured spatiotemporal factors driving our estimates of the relationship of our work organization measures and pay.

$\mathbf{x'_{i,j}}$ measures human capital controls from the ACS and LEHD. A key aspect of our study, adding these controls allows us to study the role of shifting worker selection and cumulative advantage in observed earnings changes. Workers with higher education and experience will tend to earn more. Other demographic factors may also confer advantaged labor market status on workers. Our ACS-based demographic features including White, Black, Hispanic non-Black, and Asian and other; female; education as high school or less, some college or an associate's degree, a four-year-college degree, or a graduate degree; and age in years plus age squared to incorporate non-linearity.

As the tightest measures of cumulative advantage, $\mathbf{x'_{i,j}}$ also includes 4 work history variables that we create from the LEHD earnings records, a key advantage of using longitudinally-linked data. We record all earnings for each worker in the 24-quarter time window prior to their job start quarter.[3] To capture key work history human capital factors, we record (i) mean earnings, categorized into 10 deciles for non-linearity, reflecting prior signaled produc-

[3]This series is never left-censored because the LEHD data start before 2004.

tivity and activated bargaining power; (ii) the slope on those earnings over observed quarters, categorized into ten deciles for non-linearity, reflecting prior earnings growth; (iii) a binary measure of whether the worker joined the labor market in the last 4 years, reflecting labor force choices beyond age and education; and, (iv) a binary measure of whether the worker history data include a zero-earnings quarter immediately preceding the job start quarter, possibly reflecting education, entrepreneurship, or unemployment prior to starting their new job (we exclude zero earnings quarters in the first three work history variables). This means that we systematically exclude workers' first jobs. Insofar as earnings increases simply reflect employers shifting toward new hires who were previously advantaged in the labor market, adjusting for these work history measures should reduce the earnings premium associated with job upgrading.

Although we are proposing the generality of our mechanisms across labor market segments, our project is motivated by a long line of research on sources of upward mobility for disadvantaged workers. To test for heterogeneity across broad occupation groups, and determine whether the pattern we see here holds for workers entering non-managerial/professional jobs, we additionally split our main sample into groups based on SOC code: managerial/professional (SOC groups 11-29) and non-managerial/professional (SOC groups 31-53). Though we also provide all results for the managerial/professional sample in Appendix Table A.4, we focus in our Results section on the main sample and the non-managerial/professional sample, which comprises 293,000 job starts by 277,000 workers at 15,000 employers.

We run two additional sets of analyses which are variants on this saturated model, described further in the following section: a selection analysis of how increasing job complexity and autonomy impacts the human capital composition of workers taking those jobs; and, an analysis of the long-term worker earnings outcomes of allocating more complex and autonomous tasks. A series of appendix sections and tables include additional descriptions and tests raised by our main analyses.

7 Results

7.1 Earnings Effects of Changing Work Organization

How does work organization impact earnings? Table 2, Panel A, Model 1 offers an unsaturated model, in which we predict starting earnings for all workers in our sample using task characteristics and calendar quarter fixed effects. Complex off-the-job and on-the-job and high discretion tasks are all strongly associated with higher pay. Indeed, these task categories alone have an R^2 of 0.20, or around two thirds of a standard Mincer human capital regression. Model 2 shows that the predictive power of work tasks is not simply a function of occupation: strong effects persist even after controlling for detailed, 5-digit SOC occupations (we group some SOC occupations when they are represented by only 1 group in the 500 category Census ACS OCC occupation codes). Notwithstanding the noise involved in translating work tasks into online job job posts, and survey coding those tasks, variation in earnings strongly tracks work tasks. We provide a fuller set of descriptive models in Appendix F that show how controls like industry, education and employer size relate to the effects of our novel task measures.

[Table 2 about here.]

However, differences in tasks across jobs could be largely a function of working at different types of companies, spanning different industries, production technologies and market contexts. A retail clerk at Dollar General may have different tasks than one at Trader Joe's, but there are many other contributors—business strategy, market segment, ownership structure—besides tasks that could drive wage differences between those jobs. Model 3 adds occupation by employer fixed effects, which attenuates task earnings effects by between one half and two thirds. Substantial cross-employer variation in occupations' task composition indeed track earnings. But even looking at task changes within a given employer by occupation, complex and high discretion tasks strongly predict earnings.

This within-job variation could also be a function of differences between skill demand or regulation between local labor markets, that track both earnings and task composition. For example, a Dollar General in Cambridge, MA will pay more due to labor market tightness and minimum wage rules than one in Hurtsboro, AL. To adjust for these differences in local labor market dynamics, Model 4 adds calendar quarter by commuting zone by broad occupation fixed effects.

Model 4 shows that adjusting for time-varying, occupation-specific local labor market conditions, and comparing within the same employer by occupation job, more complex and higher autonomy tasks continue to pay more. A large shift—when a job goes from no complex, off-the-job tasks to one that is entirely complex—is associated with an 11% earnings increase (calculated by exponentiating the logged earnings model). An analogous shift toward on-the-job complex tasks brings a similar earnings increase. The effect of a shift toward high discretion or autonomy tasks delivers a slightly smaller 7.5% earnings increase. Shifting work organization thus delivers higher paying jobs.

How much of this apparent within-job earnings growth reflects a shift towards increased cumulative advantage, or hiring higher skilled workers into newly higher paying jobs? Model 5 in Table 2, Panel A, adds a detailed set of controls for human capital and work history, described above: prior earnings, prior earnings trajectory, education, age, demographics, recent labor market entry, and a pre-hire non-employment spell. Adjusting for these attributes allows us to distinguish between earnings changes due to hiring different workers from those attributable to true pay premiums. Model 5 shows that the importance of these controls, and therefore of shifting worker selection, varies across task types. A full half of the apparent earnings effect of shifting toward more complex, off-the-job tasks is accounted for by adjustments for worker characteristics. In contrast, consistent with the ideas discussed above, other task changes are less affected by selection: shifting toward complex on-the-job tasks is attenuated by around one third by selection, while shifting toward high discretion tasks attenuates by less than 10%. These patterns in effects and proportion explained by

human capital, persist when, rather than preferring the quarter prior to the job start quarter, we prefer BG information from the same quarter at job start (see Table A.5).

Note that the magnitudes estimated above consider transformative changes in work organization, of a type that is unrealistic for most employers. A more modest 20 percentage point change in task dimensions—equivalent to changing a single task listed in a 5-task job post—brings a 2% earnings increase for complex tasks and a 1.5% increase for high autonomy tasks. These realistic effects are smaller, but still meaningful real earnings increases. For comparison, from the 1970s through the early 1990s, real earnings were entirely stagnant for blue collar workers (Massenkoff and Wilmers, 2023). Work reorganization offers a rare source of quantitatively meaningful earnings growth.

7.2 Shifting Composition of Workers in Response to Work Reorganization

In Table 3 we directly study how worker selection changes with these complexity and autonomy shifts, by running similar within-job task change models (i.e., with other predictors the same as Models 5 and 10 of Table 2) and swapping out quarterly earnings for worker characteristics as dependent variables. When a job shifts toward more complex, technical tasks, it draws workers who are slightly more likely to have a college degree (3 percentage point higher college share) and who have substantially higher pay at their prior job (12.6%).

Shifts toward complex on-the-job tasks bring smaller changes in prior earnings and demographic composition, consistent with the argument about commitment and on-the-job training above.

[Table 3 about here.]

Shifts toward increased autonomy, in contrast, minimal changes in selection, consistent with the earnings results in Table 2. More discretion brings little change in a job's demographic composition and a small increase in college-educated hires. Most strikingly, there is no increase in prior earnings for hires into higher discretion jobs: raising autonomy by the

same proportional amount gives workers a larger bump relative to their prior employment, compared to raising complexity.

These results demonstrate that work organization that shifts jobs toward more complex and higher discretion tasks delivers higher earnings. It also shows that these effects persist after rich controls for worker characteristics. This means that these task changes bring organization-specific earnings effects, which cannot be accounted for by labor market-wide supply and demand for generic skills. However, the relative importance of shifting worker selection varies substantially across distinct types of work reorganization. Cumulative advantage dominates increases in off-the-job or technical task complexity, while it increases less when employers shift toward more complex tasks learned on the job or increased autonomy.

Do these patterns hold for frontline workers, specifically, or do they only characterize white collar jobs? Panel B of Table 2 shows the same models in Panel A, but restricting to the subset of non-managerial and professional jobs in our sample. Overall, panel B shows that improved task allocation also raises earnings for frontline jobs. The main difference is that complex, off-the-job tasks deliver a consistently smaller earnings premium across models for frontline jobs compared to the full sample. Figure 2 compares the magnitude of within-job effects visually and confirms that the main difference across samples lies in the weaker role of complex, off-the-job tasks for non-managerial/professional jobs. Even in frontline jobs, improvements in work organization delivers earnings increases. But increased technical complexity mainly flows toward more advantaged workers, while increasing autonomy or adding more complex but tacit knowledge tasks do little to exacerbate cumulative advantage.

[Figure 2 about here.]

7.3 Long-term Effects of Changing Work Organization

While work reorganization may raise initial pay, does it also affect long-term worker trajectories? The theory proposed above implies that both autonomy and on-the-job complexity should provide durable earnings increases, not just extra pay at the time of hire. To avoid

worker resistance and bolster worker commitment, employers need to pay a consistent premium to workers.

To assess this, we construct a long-term earnings measure based on workers' average earnings four years after their job start. As mentioned above, this limits our analysis to a subset of 363,000 job starts that begin early enough in our panel that we can observe the job starter's earnings 4 years later. Tables A.1 and A.2 shows that this long-term sample is very similar to the overall sample in terms of demographic and job attributes. Note that we mitigate selection and attrition concerns by including all workers who remain employed 4 years out, regardless of whether they have persisted in the same job they were initially sampled under.

Models 1 and 2 in Table 4 correspond to within-job task change earnings models analogous to those in Table 2 Models 4 and 5, predicting earnings 4 years after the job start, rather than starting earnings. For a same-sample comparison, we also re-estimate the short-term earnings results from Table 2 for this smaller sample.

[Table 4 about here.]

The results in Table 4 show that the immediate earnings benefits of a job shifting toward more complex tasks or higher discretion persist over the long-term. Workers hired into a more complex version of the same job are on average still making 6 (off-the-job) or 8 percent (on-the-job) higher earnings 4 years after hire than are very similar workers—with the same prior pay, demographics and education level—who were hired into a lower version of the same job (Table 4, Model 2, again considering that moves of about 1/5 of these parameters are more common in our data than moves form 0 to 1). The results are even larger for jobs that shift toward higher discretion: a similar worker hired into a higher discretion version of the same job makes 10% more 4 years out.

Indeed, after adjusting for selection and worker characteristics, the point estimates for long-term effects are actually larger in the long run than the short-term benefits estimated

in Models 5 and 6. These results suggest that the earnings premiums associated with shifts in work organization offer permanent benefits to workers, rather than temporary increases for new hires. The changes we study set workers on an upward earnings path.

8 Discussion

Are there alternative forms of work organization that might interrupt cumulative advantage processes in the labor market? Are there versions of job upgrading that would not merely shift hiring towards higher-skill workers? This inquiry builds on a long tradition of research that has grappled with providing managerial and organizational solutions to macro-level problems of pay stagnation, job quality and inequality. Inspiring proposals such as high-involvement work systems (Osterman, 2006), training and learning organizations (Lynch, 1992), enabling bureaucracy (Adler and Borys, 1996), flexible specialization Piore and Sabel (1984), and job enrichment Herzberg (1968), have been difficult to study and test because of limited data tracking tasks across otherwise similar jobs for similar workers, and challenges in measuring work organization dimensions like autonomy across the enormous variety of work tasks.

We focus on a common denominator of many of these proposals by studying work organization and the allocation of tasks specifically. Work organization can affect pay either by simply increasing skill requirements, or by requiring more worker commitment or consent. These distinct mechanisms linking work reorganization to increased earnings have divergent expected effects on shifting worker selection and cumulative advantage. Increasing general skill requirements will tend to shift employers toward hiring higher skilled workers. Increasing autonomy or employer-specific skill will bring no such shift in selection, as employers seek to pay a premium relative to any workers alternative earnings opportunities. Buying loyalty while switching toward more advantaged workers simply becomes more expensive for employers. In highlighting mechanisms linking pay to resistance and commitment, our study builds on classical organizational theories of employer sources of pay differentials over and

above workers' skill level and labor market institutions (Granovetter, 1981).

To test these ideas, we develop a new data infrastructure to study the earnings and selection effects of tasks allocated along these lines, using tight comparisons among similar workers in the same jobs, but who are hired under different task allocations.

We find that job upgrading is strongly associated with increased worker earnings. Even comparing within the same occupation at the same employer, and adjusting for local labor market dynamics, shifting toward more complex and higher autonomy tasks increases starting pay. For complex tasks typically learned off the job through formal education, half of this earnings increase is attributable to shifting selection of workers into more complex jobs. But for tasks learned on-the-job, typically involving work experience and tacit knowledge, selection changes less, and for high-discretion or autonomy tasks, very little of the task-based earnings benefit is due to shifting worker selection. We also find that these benefits persist for non-managerial/professional, frontline workers and that the earnings advantages of more complex and higher autonomy tasks persist, and even increase, over the four years following task assignment. Improved work organization, and especially increased discretion, can deliver robust earnings benefits that are not explained by local labor market dynamics or shifting human capital. These effects are not temporary.

8.1 Limitations

How do the characteristics of our new data infrastructure affect the interpretation of the results? First, our research design does not allow random assignment of work reorganization events to jobs. We address selection on unobservables by studying changes within jobs, defined as occupation-employer pairs, over time. We use rich controls such as each job starter's work history, company-occupation fixed effects for time-invariant factors of employers and positions, and time-varying patterns in pay across local industries. These variables control for major alternative explanations for the relationship between our work organization dimensions and pay, such as changes in minimum wage or a legacy of unionization. This, however,

leaves out time-variant features of companies and positions.

To get something like a causal effect of work organization changes on cumulative advantage, future research could attempt to randomize autonomy and on-the-job learning in job design. Such an approach is exciting, but would likely require narrowing in on a specific company or workplace. This type of research design makes it difficult to observe new hires' prior trajectories at other workplaces, as is needed for studying cumulative advantage. Most promising would be to combine the administrative data approach used here with shocks that affect multiple organizations.

Second, while we weight our estimates to U.S. national occupational and demographic proportions, these estimates continue to rely disproportionately on workers at larger companies (about 77% of our weighted sample job starts are a companies with more than 500 workers, compared with around 50% of workers overall). As mentioned above, we control for industry and company size as well as unobserved, time-invariant company factors. But it may be that the relationship between complexity and/or autonomy and pay is different in the kinds of employers that disproportionately fall into our sample, compared with the kinds of employers that disproportionately fall out. In a more complex way, if an occupation tends to fall in smaller companies, it could conceivably have a different relationship between autonomy and pay when it is outside of its normal employer type context, perhaps relating to less job turf in larger relative to smaller employers (Wilmers, 2020). As discussed in subsection 5.5, whereas we balance occupation because of the theoretical concern about the varying composition of high autonomy and complexity tasks across occupation, it is less clear that large employers should be more responsive to job upgrading than pay at small employers. Indeed, the organizational research literature suggests greater wage rigidity in large employers, making our test conservative.

Future research could study this directly, by testing work organization mechanisms across contexts. For example, are there some organizational contexts that magnify the effectiveness in autonomy-based job upgrading boosting pay for frontline workers? A clear question in

this vein is how features of occupations interact with features of employers to shape the distributional impact of job upgrading across exposed workers. This is exciting future work, motivated by the findings we present here.

Third, the fact that we observe employer reports of future tasks, rather than records of performed tasks, suggests future research where workers and managers carefully keep track of the tasks performed by each worker. Job posts are projections made by hiring managers. From a simple perspective, employers should seek to accurately define jobs to maximize candidate fit. However, the organizational literature suggests that hiring managers, due to generosity, apathy, or status pursuit, may incorporate more complex or autonomous tasks into job posts than are actually performed as a justification for or alongside higher pay. Such behavior would positively bias our estimates even in the absence of real, shop floor changes in work organization. If, on the other hand, employers tend to project higher discretion tasks than what actually is performed, and workers are paid the same regardless of this over-projection, job posting-based measures will be biased toward zero. Our within-firm-occupation approach means that any problematic measurement error here would have have to be within-job, rather than some consistent feature of an industry or firm's job advertising strategy.

Again, studies randomizing task allocation in real jobs could address this scope condition, but that approach will likely forsake the breadth we obtain here. One promising direction could be augmenting the The Bureau of Labor Statistics American Time Use Survey, which uses a time diary instrument to ask detailed questions about respondents' leisure activities, with direct questions about work tasks to a time use survey of that type could usefully address the problems involved in using job posts to measure tasks (Autor and Handel, 2013). It could be useful to estimate whether and when performed tasks or our work organization dimension deviate from tasks in job post data, and we share our survey results at [TK: after review] to foster this type of work.

A final limitation is that we are unable to assess how our results would change if we

focused on the more restrictive condition of employer by job-title fixed effects to proxy for job positions, rather than employer by occupation fixed effects. Our ACS data does not include job title, so it is impossible to compare these analyses. Part of the employer by occupation variation we use is no doubt due to within-occupation, but between-job title comparisons. This approach has some advantages. Some job title changes are merely ceremonial reframings of similar tasks. Other job title changes demarcate exactly the work reorganization processes we study. If a 'front desk clerk' becomes a 'patient care coordinator' because insurance coordination was a task added to check-in and answering phones, then comparing only within-job title would mean dropping a whole array of job upgrading examples.

However, there will clearly be instances where two consistent job titles appear in an organization, that are hired within one occupation. Consider the familiar example of professors at universities. Occupation codes carefully distinguish faculty by discipline, differentiating business professors from economics professors, for example. But, the codes do not distinguish seniority-based differences like assistant and associate ranks. As such, task differences between those ranks could enter our estimates of job upgrading. Future work could use mobility data attached to job title, for example from LinkedIn, to examine how to better group the many job titles in job posting data, into a level of "reasonable alternative jobs" for each worker.

8.2 Conclusions

Notwithstanding these limitations, our study makes several contributions. First, we show that, across many companies and occupations, organization-level task allocation decisions affect cumulative advantage. Within-labor market and within-occupation, we show substantial variation in task allocations that accounts for variation in earnings and worker characteristics. But studying changes in work organization over time, we find substantial heterogeneity in how work organization and pay changes affects workers selection. When employers increase skill requirements by adding complex tasks typically learned off the job, pay increases

but largely by hiring workers who were already in higher paying jobs. In contrast, when employers increase autonomy or complex tasks learned on the job, worker selection changes less. The specific channels through which work organization affects pay are critical for shaping cumulative advantage responses.

Second, we show how insights from organizational case studies can be scaled up to study many occupations and companies across the economy. Discussions of task-driven learning and of workplace control are common in organizational ethnographies and case studies of tasks and work organization in specific occupations and workplaces. But these rich insights are rarely tested in economy-wide data. Moreover, we build on earlier theory to present these mechanisms as functions of worker commitment and resistance. These contributions motivate further research on what drives variation in how employers organize work.

Third, for policymakers interested in improving labor market outcomes for disadvantaged workers, our project holds several insights. First, it is insufficient to advocate for higher pay per se: some bases for earnings increases shift hiring toward more advantaged workers. Job upgrading can be a form of credentialing that actually diminishes opportunities for the worst-off workers. Second, however, in distinguishing types of work reorganization, we find that changes that increase autonomy and on-the-job learning can provide earnings premiums with minimal selection changes. Federal industrial policy could focus on encouraging jobs with these features, rather than on encouraging industries with high wages alone. At the local level, economic development agencies could work with employers to shift toward these job designs that provide opportunity for disadvantaged workers. Long-term, policymakers can encourage the technologies and organizational innovations that support worker commitment and magnify the value of worker autonomy.

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Table 1: Characteristics of Employers Offering Complex and Autonomous Front-Line Tasks

Variable	Sample Mean	Off	On	Discretion	Relationships
Employment	1150	283	483	-397	199
Logged Employment	4.99	0.32	0.24	-0.04	-0.04
Logged Payroll	8.79	0.54	0.33	0.05	-0.10
Logged Revenue	9.39	0.36	0.46	-0.07	-0.20
Employment Growth	0.046	-0.004	-0.016	0.022	0.005
Revenue Growth	0.265	0.015	-0.021	0.037	0.027
Revenue/Employee	162.0	14.7	37.1	-2.1	-25.0
Profits/Employee	106.0	0.6	34.5	-10.8	-22.1
Pay/Employee	55.9	14.1	2.6	8.7	-2.9
Mean lnQ2 Pay	8.96	0.18	0.08	0.08	-0.04
Variance lnQ2 Pay	0.23	0.02	0.00	0.01	0.01
Founded Pre-1980	0.29	-0.02	0.03	-0.05	-0.03
Founded 1980-1999	0.31	0.00	0.00	0.00	-0.02
Founded Post-2000	0.40	0.02	-0.03	0.05	0.04
C Corporation	0.29	0.05	0.06	0.01	-0.03
S-Corp/Partnership	0.56	-0.13	0.00	-0.03	0.00
Non-Profit	0.15	0.08	-0.06	0.01	0.03
Manufacturing+	0.26	-0.06	0.09	-0.06	-0.09
Retail+	0.15	-0.08	0.04	-0.02	0.00
Business Services+	0.28	0.10	-0.01	0.10	-0.01
Health+	0.21	0.07	-0.09	-0.08	0.10
Education+	0.11	-0.03	-0.03	0.05	0.00

The sample used in this table comprises 59,000 employers. Columns labels with our work organization scores reflect the difference in average value between employers that are above versus below the median of the focal work organization dimension. 'Off' and 'On' refer to our off-the-job complexity and on-the-job complexity measures, respectively. The predictors were created using Burning Glass data. Mean second quarter pay and variance in second quarter pay, were created using the U.S. Census Longitudinal Employer-Dynamics Database; and all other variables were created using the Census Longitudinal Business Database.

Table 2: How Work Organization Affects Cumulative Advantage

Sample	A. Main				
Model	(1)	(2)	(3)	(4)	(5)
Complexity, Off	1.56***	0.450***	0.123***	0.104***	0.054***
	(0.034)	(0.025)	(0.015)	(0.013)	(0.011)
Complexity, On	1.65***	0.344***	0.120***	0.091***	0.064***
	(0.043)	(0.033)	(0.016)	(0.015)	(0.012)
Discretion	0.750***	0.145***	0.088***	0.072***	0.066***
	(0.056)	(0.033)	(0.019)	(0.016)	(0.013)
Relationships	-0.350***	-0.105***	0.016	0.011	0.010
	(0.043)	(0.025)	(0.014)	(0.013)	(0.011)
Start Quarter	×	×	×		
SOC5		×			
SOC5∧Company			×	×	×
StartQ∧CZ∧SOC2				×	×
Human Capital					×
R^2	0.20	0.47	0.67	0.73	0.82
Within R^2	0.18	0.01	0.00	0.00	0.33

Sample	B. Not Managerial/Professional				
Model	(6)	(7)	(8)	(9)	(10)
Complexity, Off	0.933***	0.490***	0.087***	0.084***	0.035*
	(0.043)	(0.037)	(0.020)	(0.019)	(0.015)
Complexity, On	1.25***	0.303***	0.126***	0.090***	0.065***
	(0.045)	(0.039)	(0.018)	(0.016)	(0.014)
Discretion	0.404***	0.102*	0.086***	0.065***	0.062***
	(0.069)	(0.043)	(0.023)	(0.019)	(0.016)
Relationships	-0.561***	-0.151***	0.014	0.004	0.011
	(0.046)	(0.031)	(0.016)	(0.015)	(0.013)
Start Quarter	×	×	×		
SOC5		×			
SOC5∧Company			×	×	×
StartQ∧CZ∧SOC2				×	×
Human Capital					×
R^2	0.17	0.38	0.63	0.71	0.79
Within R^2	0.14	0.02	0.00	0.00	0.27

Significance: * $p < .05$; ** $p < .01$; *** $p < .001$ (two-tailed tests). The dependent variable is second quarter pay throughout. The main sample (Panel A) comprises 611,000 job starts by 570,000 workers at 23,500 employers and the non-managerial/professional sample (Panel B) comprises 293,000 job starts by 277,000 workers at 15,000 employers. We created this table using data from Burning Glass and the U.S. Census Longitudinal Employer-Dynamics Data linked to the American Community Survey.

Table 3: Studying How Work Reorganization Changes Worker Selection

DV	4-Year College	Female	Bl. or His.	Age
Model	(1)	(2)	(3)	(4)
Complexity, Off	0.031***	-0.032***	-0.026**	0.109
	(0.007)	(0.010)	(0.010)	(0.345)
Complexity, On	-0.002	-0.056***	-0.014	0.039
	(0.008)	(0.013)	(0.012)	(0.454)
Discretion	0.033***	-0.008	0.000	-0.810
	(0.010)	(0.012)	(0.013)	(0.437)
Relationships	0.012	0.007	-0.009	0.044
	(0.008)	(0.010)	(0.010)	(0.331)
SOC5∧Company	×	×	×	×
StartQ∧CZ∧SOC2	×	×	×	×
R^2	0.62	0.53	0.44	0.47

DV	WH Pay	WH Slope	WH Gap
Model	(5)	(6)	(7)
Complexity, Off	0.119***	0.000	-0.013
	(0.020)	(0.005)	(0.010)
Complexity, On	0.085***	-0.006	0.007
	(0.022)	(0.006)	(0.012)
Discretion	-0.005	-0.001	-0.005
	(0.025)	(0.005)	(0.013)
Relationships	0.000	0.000	0.000
	(0.019)	(0.004)	(0.009)
SOC5∧Company	×	×	×
StartQ∧CZ∧SOC2	×	×	×
R^2	0.57	0.33	0.37

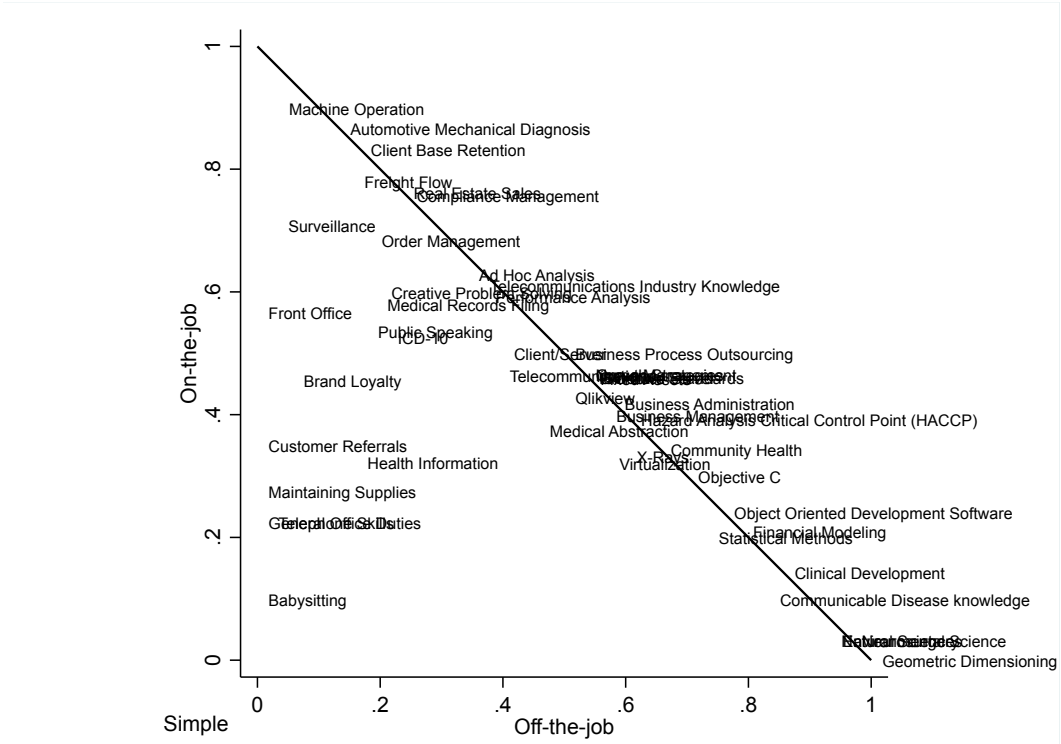
Significance: * $p < .05$; ** $p < .01$; *** $p < .001$ (two-tailed tests). The dependent variables, listed across the top, are produced using the human capital controls used in our main models, with 'WH' standing for work history. We use the same sample as in our main models, comprising 611,000 job starts by 570,000 workers at 23,500 employers. We created this table using data from Burning Glass and the U.S. Census Longitudinal Employer-Dynamics Data linked to the American Community Survey.

Table 4: How Work Organization Affects Long-Term Pay

DV	4-Year-Later Earnings		Quarter 2 Earnings	
Model	(1)	(2)	(3)	(4)
Complexity, Off	0.116***	0.061**	0.091***	0.045**
	(0.025)	(0.022)	(0.017)	(0.014)
Complexity, On	0.107***	0.080**	0.071***	0.048**
	(0.031)	(0.028)	(0.020)	(0.016)
Discretion	0.107**	0.094**	0.058**	0.051**
	(0.033)	(0.030)	(0.0208)	(0.017)
Relationships	0.049*	0.047*	0.014	0.009
	(0.025)	(0.023)	(0.017)	(0.014)
SOC5∧Company	×	×	×	×
StartQ∧CZ∧SOC2	×	×	×	×
Human Capital		×		×
R^2	0.57	0.63	0.74	0.83

Significance: * $p < .05$; ** $p < .01$; *** $p < .001$ (two-tailed tests). For dependent variables, *4-Year-Later Earnings* reflects the average earnings 16 and 17 quarters after job start and *Quarter 2 Earnings* is the same dependent variable as in the main results. The sample used in this table comprises 363,000 job starts by 343,000 workers at 15,500 employers. This table draws on using data from Burning Glass and the U.S. Census Longitudinal Employer-Dynamics Data linked to the American Community Survey.

Figure 1: Diagram of Survey Results on Location of BG Tasks on Complexity and Autonomy Work Organization Dimensions



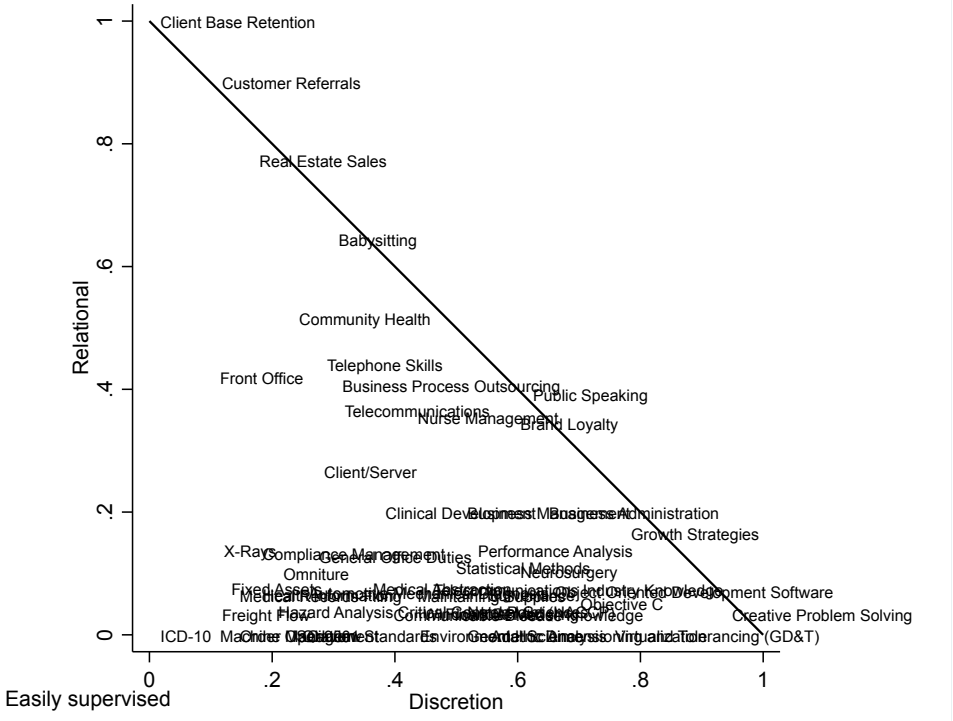
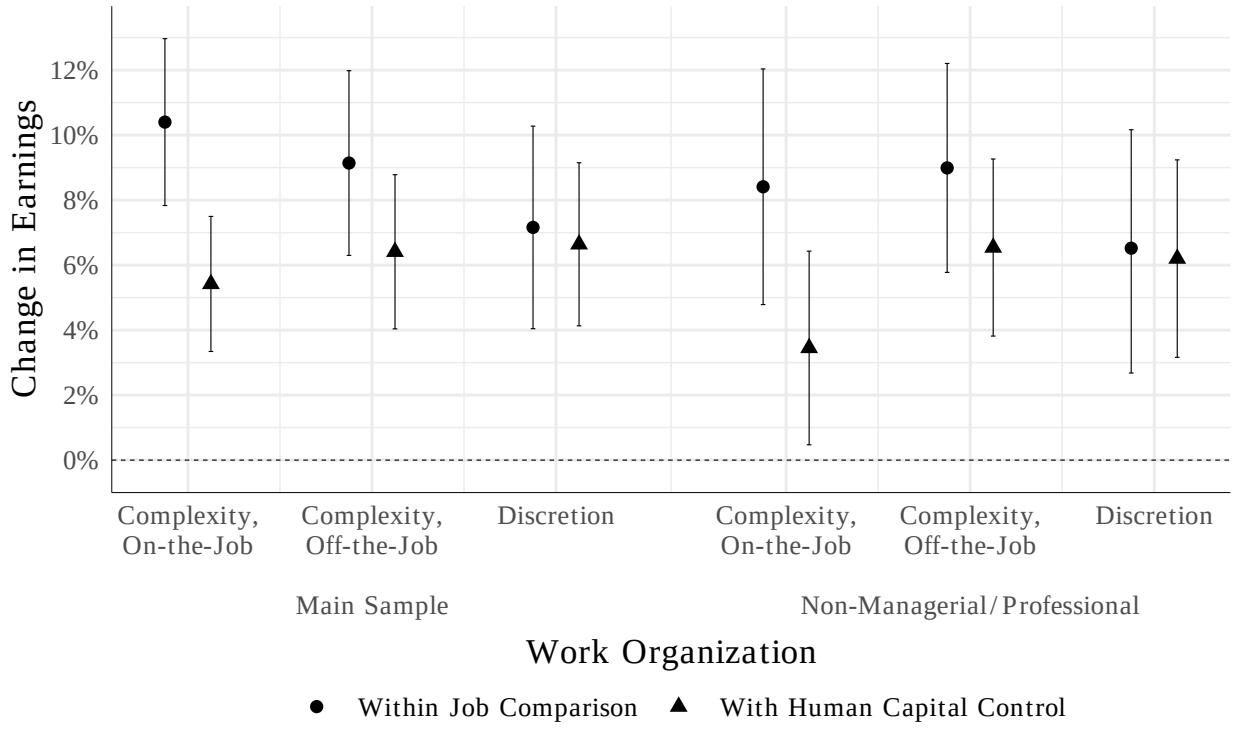


Figure 2: How Work Organization Affects Pay, Plotted by Dimension, Sample, and Model



A Survey Methodology

We conducted a series of online surveys to categorize tasks by the degree of autonomy and types of complexity involved. These surveys were completed from October 14th through October 21st, 2021, using Qualtrics forms distributed via the Prolific platform.

We fielded two surveys of a similar format: one focused on task autonomy, and the other gauging task complexity. Participants self-selected into these surveys, based on their professions, and were asked to rate a series of tasks associated with their occupational fields (we fielded multiple versions of each of the surveys to obtain full coverage of the tasks for that occupational field). In every survey type, tasks were randomly and evenly assigned across participants, with an average of 30 responses per task. Figures A.1 and A.2 give examples of the instructions given to workers with the concept definitions presented to them. We occasionally allowed people to participate in adjacent fields if the volume of available respondents in that field was low.

[Figure 3 about here.]

[Figure 4 about here.]

We also placed several restrictions on the participant pool, consistent across all surveys. We required respondents be aged 30 to 95, in order to prevent a possible skew after a viral TikTok video caused Prolific to temporarily have a disproportion of young adults in the participant pool. We also required all participants to reside in the United States and have a non-Prolific full-time job. Our average wage goal was $15 per hour, and respondents' average time ranged between 4 to 11 minutes, depending on the survey.

The surveys followed a specific and consistent flow across occupational fields. Participants were asked to pick their occupational fields (roughly corresponding to 2-digit NAICS codes) and industries from multiple choices (these options were selected from the Bureau of Labor Statistics' NAICS codes), and declare demographics. Age options were provided as exact

numbers ranging from 20-80 years old; even though our Prolific pre-screening excluded people below 30 years, we included ages 20-29 as an option to validate whether our participants were actually 30 years or older. Analyzing demographics of the respondents also allowed us to ensure our data was not affected by the viral TikTok video.

The next section of the survey was a training section with interactive feedback, given after the participant submitted their answers. Figures A.1 and A.2 reproduce the prompt provided in this section. The training included instructions for categorizing six tasks, which were consistent across all surveys within each dimension. Specifically, in the discretion survey, participants were asked to choose whether a task required high discretion, high rules or supervision, or significant interactions with coworkers and clients. For the complexity surveys, participants were asked whether the specific task was simple, complex and learned on the job, or complex and learned off the job. In every case, the respondents also had the option "I don't know."

The main body of the survey showed each participant 50 different tasks based on the prevalence of those tasks across occupational fields. Tasks were randomly and evenly distributed across each respondent for consistency. Additionally, to ensure each participant was engaged with the survey and actively thinking about each task, we separated the tasks into 10 per page, for a total of 5 pages, and included an attention check question. At the end of the survey, we also included a feedback box for respondents to leave any comments on the tasks and/or the survey.

To construct the questions and instructions, we utilized the top 2,000 most frequently referenced tasks from Burning Glass, and sorted them based on the most frequently associated SOC code. We then constructed a crosswalk between SOC codes and Prolific occupation categories, and sorted the tasks based on the Prolific categories. This meant that some occupations received more tasks than others, and to ensure each task was evenly represented, we (1) rounded each survey to the next multiple of 50 tasks by using the top most common BG tasks (for example, if a occupation category had 84 tasks, we extracted 16 additional

ones from the top most common tasks in Burning Glass, without repeating any values from the original occupation's tasks, to create a list of 100 tasks), and (2) matched the proportion of respondents requested to the number of tasks in each occupation's survey. This created even coverage across the surveys, ensuring that each participant saw 50 tasks, and that we received 30 responses per task. The number of participants per occupation category was determined based on the frequency of the occupation category, based on the Burning Glass frequency list.

B LEHD Data Restrictions

We identify the worker, employer, and start quarter for hundreds of millions of jobs from the first quarter of 2010Q1 through the first quarter of 2022. For privacy protection purposes, our LEHD coverage is limited to a random subset of U.S. states, as are all other external researcher Census projects. We can observe LEHD data for 24 U.S. states representing 50% of U.S. employment across the 5 main U.S. labor market regions: AZ, CO, DE, HI, IL, KS, MN, MD, NB, NV, NM, NY, ND, OH, OK, PA, TN, TX, UT, VA, WA, DC, WI, and WY.

We essentially follow an established LEHD workflow (Dunne et al., 2004) to identify job starts in the LEHD: we (i) export the LEHD Employment History Files quarterly earnings data from 2004Q1-2022Q1, using years prior to 2010 only for the work history controls discussed below; and, (ii) merge the state-level employer IDs in the earnings data to the main Census employer IDs using the Employer Characteristics Title 26 crosswalk, because the Census IDs are associated with employer names used in our BG-LEHD merge. We then (iii) group about 1% of employers based on shared federal Employer Identification Number (EINs) that are not associated with any larger entity in one year but are in another, to trace smaller employers moving between the Census's core classification of one- versus multiple-workplace employers over time; (iv) adjust to 2021 dollars using the Bureau of Labor Statistics Consumer Price Index; and, (v) keep only the first job start per worker-employer that meets these criteria. This allows workers to have multiple job starts in our

study so long as they are with different employers.

We follow prior literature using the LEHD to trace substantial employment relationships by excluding small and transitory payments, and to reduce the impact of large changes in hours worked on our results (Haltiwanger and Spletzer, 2021). We require, in 2021-adjusted dollars, (vi) $7,552 in earnings, or the equivalent of a full-year working 20 hours per week at the federal minimum wage, over the length of the job or the first 2 years of the job, whichever comes first; (vii) $1885 in second quarter earnings, or the equivalent of 20 hours per week at the federal minimum wage; (vii) third quarter non-zero earnings, reducing the influence of partial quarter job spells, i.e., weeks worked, on our main dependent variable of second quarter earnings. This filtering of short employment relationships is common in work using the LEHD and other administrative earnings data (Dunne et al., 2004). We also exclude a sliver of jobs where our identified second quarter earnings are not the majority of observed earnings for the worker in that quarter. We log all earnings outcomes throughout.

C Employer Name Merge

Our goal is to link employers identified in the Burning Glass data with the U.S. Census administrative records of all businesses. We iteratively developed a three-step strategy to incorporate employer into our merge novel of the BG and LEHD data: cleaning, fuzzy matching, and processing the employer-occupation-CZ units.

Three key challenges influenced our name matching: (i) BG employer names tend to be oriented towards employees seeking jobs, whereas the Census names tend to be oriented towards administrative records; (ii) the Census Bureau business data include both a primary and secondary names for each *workplace*, where one of the names is often not useful for matching, but either name may contain the business name as it would be conceived by BG, and either may contain common words that produce false matches to BG data; and (iii) the BG data are more disaggregated than the Census data when business units post jobs under their own name, and more aggregated when franchises are grouped by BG into one

"employer," whereas the corporate name appears only in the secondary name column in the Census data.

For the (A) cleaning step, we clean all names through standard actions such as upper-casing, removing punctuation, and removing tags such as "Incorporated." Sometimes these tags are useful for identifying the true employer name, for example "headquarters," but often they are not, instead capturing something like the city name that the workplace exists in or the name of the managing partner in a small firm. We apply a host of Census-data-specific cleaning strategies based on an iterative review of the cleaned data. For example, we develop an algorithm to clean out common words that are used to mark specific workplaces within employers. We then create pseudo-Census-employer-IDs for franchises that otherwise have distinct IDs in the Census data, effectively matching many of these entities by hand. One reason we want to do this is that we then categorize each employer in the BG and LEHD data based on whether its posts or starts were largely in only one state or across multiple commuting zones (CZs), to use as a merge criterion. While the vast majority of employers post and hire in a single CZ, the majority of workers are in multi-CZ employers.

For the (B) fuzzy matching step, we first set aside all employer names with fewer than 6 characters, which are only allowed to exact match. We then loop through all CZs as well as an indicator for multi-CZ, looping again through all sets of first 3 characters that appear within each CZ on both the BG and LEHD side. We create a name similarity score based on cosine distance for all names, after extensive testing of whether this assumption excludes a meaningful number of businesses that otherwise could be matched. We then only keep name matches that are either exact matches or that match with greater than 66% similarity and obtain 1 of 11 criteria that we defined through hand review with false match rate testing. For example, 1 criterion is that both names share the first two words and the third work on either or both side is a common ending word. This work creates a list of all employer names that match across the two databases, but this is not always a 1-1 list and may include chaining, for example when name A links to name B that links to name C, etc.

For the (C) processing the employer-occupation-CZ units step, we employer-occupation-CZ units based on clusters on the bimodal network of all sets of Census and BG employer names that fulfill our occupation and CZ criteria. We do this rather than creating employer-level units to avoid the issue of common names producing large clusters of associated names, i.e., "American Distributors" linking to "American Distribution" and to "American Tire Distributors" when there entities in fact share no occupational or geographic overlap between the job post and start data. Units without occupational or geographic overlap would be deleted anyway in our merge, and, in our initial tests, led to false job to post linking due to large clusters of names. We then conduct hand testing of matched entities. On the one hand, we remove some BG-LEHD links that appear to be false clusters even when they pass our 12 match criteria. On the other hand, we have a research assistant hand match the 1,000 largest unmatched BG entities.

Note that, in the next analytic step, our study requires a job start and post to match in a narrow time window to be included in the datasets. Thus, companies that never hire in the same occupation can be "name matched," but only those that additionally have false occupation-CZ-time matches will contribute noise (i.e., attenuation bias) to our regression.

D How the Regression Weights Shape Key Sample Features

Table A.1 compares our main sample before and after weighting based on the 22 2-digit SOC occupation groups (we aggregate occupations in this table into 4 groups for clarity), and demographics, to U.S. national proportions (see subsection 5.5). Column 1 describes the Burning Glass job post data. Column 2 presents nationally representative figures for the U.S. workforce (not just job starters) based on the U.S. Census ACS survey (Column 6 presents the equivalent values based on the nationally-representative CPS-ASEC survey, based on employed workers not in school, aged 20-60, and based on the 2010-2021 surveys with person weights: We present statistics for the CPS-ASEC because, unlike the ACS, it asks works about their employer's size). Column 3 reflects our main data before weighting,

Column 4 our main data after weighting, and Column 5 our main data with the same weights but keeping only workers who start early enough in the panel to have an observable earnings measure four years after starting. Overall, Table A.1 provides evidence that our ACS-based weights erase the otherwise substantial occupational and educational imbalance in our unweighted sample: Column 4 (and 5) is much closer to Column 2 (and 6), than Column 3 is to Column 2.

Under *Occupation Group*, we see that the occupational imbalance in our unweighted main sample (Column 3) reflects the occupational distribution of Burning Glass job ads (Column 1). Managerial/professional occupations represent 50% of BG posts and unweighted sample job starts, but only 38% of workers according to the ACS and 41% of workers according to the CPS-ASEC. As we discuss above, this is concerning because the relationship between, for example, technical tasks and pay may be different among different occupations, even after using the natural log to study proportional changes in earnings. Weighting brings the "managerial and professional" proportion down to 34% for the main sample, i.e., we slightly overshoot our goal due to correlations with the demographic groups we want to upweight. Regarding key demographics, our unweighted sample is biased towards higher-educated compared with lower-educated workers, and, less dramatically, female, younger, and White workers. Weighting closes the large gap on education and eliminates the demogrpahic imbalance, but slightly increases the small overrepresentation of White workers.

Moving to *Sector*, not used in weighting, Table A.1 shows that weighting reduces some of BG's sectoral bias, though our sample still overrepresents healthcare at the expense of manufacturing. Burning Glass overrepresents Business Services^{+} (grouped here with information and finance), Health (NAICS 62 only), and, to some extent, Retail^{+} (grouped here with wholesale and transportation), while underrepresenting Manufacturing^{+} (grouped here with goods, utilities, and waste collection) and Social Services^{+} (grouped here with education and art). Note that the sector figures for Column 1 are derived from the 80% of our qualifying BG posts that include employer industry, which is not a requirement for being

available to match to a job start in our study. The CPS-ASEC distribution mirrors the ACS distribution. The pattern in BG job posts is repeated in our unweighted main sample: healthcare is overrepresented and manufacturing underrepresented. The even greater bias in our unweighted data towards health, relative to the BG posts, could arise, for example, if standardization of hospital names allows for a higher match rate between the BG and Census data. Weighting shifts our main sample towards national proportions in Retail^{+}, Social Services^{+}, Manufacturing^{+}, and to some extent in Health, though, as mentioned above, we are still substantially off from national proportions. We discuss implications for the generalizability of our study in subsection 5.5.

Weighting does little to address the overrepresentation of large employers in our study. We here baseline with the CPS-ASEC question: "Counting all locations where this employer operates, what is the total number of persons who work for your employer?" We find that 46% of workers report working in a large (500+ workers) employer. This is fairly similar to the 2017 Economic Census' estimate of 53% of employment in large employers, perhaps reflecting workers' greater tendency to consider business units as "employers," versus the Census' (and our) focus on the ultimate controlling company. Whichever figure is more reasonable, our weighted sample overrepresents job starts in large employers, at 77% of our sample. Judging from the BG figure alone (estimated from calculating the average rate of posts to the GlassDoor employment variable matched on employer name) of 62%, about 33-50% of our overrepresentation of large employers is explained by BG's overrepresentation of large employers, and the other 67-50% is due to study design features including our hand matching of large employers and the requirement that each employer have at least one CZ-occupation with two ACS-matched job starts matched to job posts in the acceptable time frames. We again discuss implications for generalizability in subsection 5.5.

E Descriptive Statistics on Key Variables

Table A.2 presents descriptive statistics on our key variables. Panel A shows that weighting reduces the rate of off-the-job complexity, consistent with down-weighting higher-educated job starters, but has little impact on our other work organization dimensions. Off-the-job complexity declines by 17%, mostly made up by the Simple reference dimension. This conforms to the over-representation of managerial and professional occupations, discussed above. While our two complexity measures are below the Burning Glass averages even before weighting, our measure of discretion is similar in Burning Glass and both our unweighted and weighted samples. This suggests that discretion is less limited to higher-educated and managerial/professional workers than one might first assume, according to the respondents from our original job task survey. This could be because higher-skill jobs tend to be more constrained by social relationships, the second reference category for this variable. More generally, the relatively high values in our BG data and our main sample demonstrates that our survey respondents rated many of the common tasks in the U.S. work force as complex and/or non easily supervisable.

Panel B of Table A.2 shows that weighting reduces the average pay and pre-start upward mobility of our sample. This is again consistent with re-balancing away from high-educated managerial and professional workers. Our main outcome measure of average second quarter pay is 13% lower for the weighted relative to our unweighted sample. That both of these measures (9.2 and 9.075) are lower than the national full-time worker average of 9.4, calculated as one quarter of the average annual pay in the ACS (which is also top-coded at around 1,000,000 annually), suggests the inclusion in our sample of some part-time workers. That weighting reduces work history earnings slope from 7.3 to 5.8%, but not work history average earnings, suggests that the up-weighted workers may be on less-upwardly-mobile trajectories than our relatively down-weighted workers – a critical feature of our study that prior survey-based and occupation-level work cannot take into account. Workers in our sample are receiving a 2-3% pay boost on average in their move, suggesting some lateral as

well as pay-oriented job moves. We estimate that 14% of workers in our sample recently joined the labor force, and 20% have recently experienced at least 1 0-earnings quarter despite possessing substantial work history, which we use as a proxy for extended leave or unemployment.

F Augmenting Earnings Models with Work Organization Measures

We run a series descriptive models to study how controls like industry, education and employer size relate to the uncontrolled effects of our novel task measures. These results demonstrate that our work organization measures pick up variation both explained by and beyond occupation and other major factors hypothesized in the existing literature to induce a correlation in these variables. This would not be the case if differences in job posts within organizations were human resource jargon not reflecting factors that predict pay, if our long analytic chain produced noise that attenuated our effects away, there similarly is no measurable effect of task reallocation on these measures after selecting an occupation, or human capital selection was driving the measured effects.

The full form of these descriptive models is:

$$y_{i,j} = \mathbf{u_j'}\beta + \alpha_{Start_Q} + \alpha_{Occ} + \alpha_{Naics} + \alpha_{Size} + \mathbf{x_{i,j}'}\lambda + \epsilon_{i,j} \tag{2}$$

where $y_{i,j}$ measures second quarter earnings for worker i at job j. We include the α_{Start_Q} start quarter fixed effect to remove the overall time trend. The parameters of interest β's, human capital controls $\mathbf{x_{i,j}'}\lambda$, and residual term $\epsilon_{i,j}$ are the same as above. We weight these regressions by our ACS-derived job-start-level factor and cluster standard errors at the employer level.

Even if a baseline correlation reveals higher pay for jobs with higher-level tasks on our key dimensions, the most obvious concern is that merely reflects a tendency to allocate more or less complex or autonomous tasks to different occupations. We address this concern by

controlling for the SOC 5-digit occupations.

employer industry (α_{Naics}) and size (α_{Size}) may also induce a correlation in our measures of task allocation and pay. Younger and more technological industries may be linked to patterns of worker technical savvy or social skill, whereas industries that emerged in the more distant past, may sit at a local equilibrium where certain management styles persist even if a reorganization of tasks and pay could theoretically improve net profitability.[4] But even controlling for industry, larger employers may pay workers more and offer more complex or autonomous jobs because of greater profit-sharing with workers or greater likelihood of organization bargaining (Cobb and Lin, 2017). These industry and employer size fixed effects afford comparison of the task-pay relationship for workers at employers of similar size and in similar product markets.

We would expect initial associations of our work organization measures and pay to disappear under the following 4 conditions: (a) subtle changes in task content within the categories defined by our controls reflect the noise of human resource jargon; (b) noise emerging from our long analytic chain - machine-learning tasks, task dimension survey, aggregating to jobs, fuzzy employer matching - attenuates any signal in the task data not reflected by characteristics such as occupation; (c) the only way to induce substantial changes in pay is to reassign workers across occupations rather than changing the tasks assigned to similarly positioned workers; or, (d) any initial positive correlations are explained by stronger workers selecting into more complex or autonomous jobs, or the kinds of employers offering more complex or autonomous jobs.

Table A.3 reports the results of our descriptive regressions, suggesting that these four conditions do not obtain. In Panel A, estimated using our main sample, we compare results from a simple model including our task scores (Model 1) with a model including those task scores as well as human capital "Mincer" controls bolstered with work history measures

[4]We measure the highest employing 5-digit NAICS industry for each employer in our sample over the 2010-2020 period based on the Census LBD data. We break employer size into 4 categories: 1-499, 500-14,999, 15,000-50,000, and 50,000+ employees.

(Model 2). In our baseline, off-the-job and on-the-job complexity are associated with more than a 150% rise in pay, discretion is associated with a roughly 75% rise in earnings, and ships is associated with a 35% decline in earnings. Alone, these complexity and autonomy factors explain a substantial amount of the variance in logged earnings in the U.S. workforce, with an R^2 measure of 18%. Adding our human capital control set of education, demographics, and work history, increases the explanatory power of our model to 63%. We see that our human capital measures explain about 30% of the baseline off-the-job complexity correlation, 35% of the on-the-job complexity correlation, 90% of the naive discretion effect, and only a small percentage of the ships association.

The results from Models 3 and 4 of Table A.3, which substitute occupation plus employer factors for the human capital controls in Model 2, suggest that occupation explains more of the baseline complexity task allocation effect, whereas human capital explains more of the baseline discretion effect. After controlling for occupation, moving off-the-job complexity from 0 to 1 increases pay by roughly 45%, with 34% and 15% for on-the-job complexity and discretion, respectively. Adding, in Model 4, 5-digit NAICS industry and employer size category alongside occupation, cuts our Model 2 human capital complexity estimates in half while still maintaining a relatively higher discretion estimate of an 11% rise in pay.

A comparison of the baseline Model 1 with the full form Model 6 of Table A.3 suggests that although our complexity and autonomy measures are highly correlated with human capital, occupation, and employer industry and size, the effects of our main predictors on pay are still substantively and statistically significant. Here, moving from none to total off-the-job complexity, on-the-job complexity, and discretion, is associated with a 13%, 16%, and 7% increase in pay. We therefore find evidence that the four conditions described above - BG tasks provide no signal, our analytic chain obscures that signal, only occupation changes have effects, and the effect is all selection - do not appear to obtain here. The results in Panel B of Table A.3, based on the non-managerial/professional jobs, confirms the relationships estimated with the main sample.

[Table 5 about here.]

[Table 6 about here.]

[Table 7 about here.]

[Table 8 about here.]

[Table 9 about here.]

[Table 10 about here.]

Table A.1: Descriptive Statistics on the Main Sample Before and After ACS Weighting

	Burning Glass	ACS Survey	Unweighted Sample	Weighted Sample	Long-term Sample	CPS-ASEC Survey
	(1)	(2)	(3)	(4)	(5)	(6)
Occupation Group						
Managerial/Professional	0.511	0.379	0.520	0.342	0.356	0.405
Sales/Administrative	0.234	0.232	0.252	0.242	0.255	0.213
Production	0.135	0.214	0.104	0.224	0.210	0.220
Services	0.120	0.175	0.125	0.192	0.179	0.162
High School or Less	·	0.401	0.200	0.364	0.347	0.349
Some College/Associates	·	0.257	0.321	0.328	0.334	0.269
Four-Year College	·	0.217	0.302	0.190	0.197	0.248
Graduate School	·	0.124	0.177	0.119	0.122	0.134
Female	·	0.485	0.540	0.486	0.491	0.465
Mean Age	·	40.7	37.2	43.1	42.3	40.8
White	·	0.627	0.671	0.698	0.702	0.634
Black	·	0.116	0.123	0.094	0.093	0.125
Hispanic	·	0.169	0.119	0.137	0.135	0.154
Asian (and other)	·	0.087	0.087	0.071	0.070	0.087
Sector (not used in weighting)						
Manufacturing+	0.113	0.190†	0.079	0.089	0.070	0.209
Retail+	0.202	0.185	0.281	0.221	0.222	0.174
Business Services+	0.271	0.198	0.239	0.230	0.247	0.217
Health+	0.198	0.135	0.270	0.261	0.274	0.138
Social Services+	0.215	0.292	0.132	0.199	0.188	0.263
Large Employer (not used in weighting)						
500+ workers	0.619‡	·	0.793	0.772	0.779	0.462

This table describes ACS national demographic proportions, our sample before and after applying ACS weighting, and our long-term sample after applying ACS weighting. All figures in this table are in proportions of the sample. † We calculate these proportions using publicly-available ACS data from IPUMS, applying person weights. ‡ We calculate mean employer size on a per-employee perspective in Burning Glass by taking the employee to post rate (.14 posts on average per employee over our period) among 500,000+ employers linked on name between the Burning Glass (which lacks employment) and GlassDoor data (which estimates employment), and weighting to a "per employee" figure. We created this table using data from Burning Glass, the U.S. Census Longitudinal Employer-Dynamics Data, the American Community Survey, and IPUMS public-use data.

Table A.2: How Weighting Impacts the Independent, Dependent, and Work History Control Variable Means

A. Independent Variable Mean Scores				
	Unweighted Sample	Weighted Sample	Long-term Sample	Burning Glass
Off-the-job Complexity	0.309	0.256	0.265	0.420
On-the-job Complexity	0.409	0.419	0.420	0.483
Discretion	0.363	0.360	0.360	0.365
Relationships	0.280	0.259	0.262	0.402
B. Earnings Variables from the LEHD				
	Unweighted Sample	Weighted Sample	Long-term Sample	
Main Dependent Variable				
Mean Second Quarter Pay	9.200	9.075	9.377	
Work History Controls				
Mean Earnings	9.039	9.044	9.079	
Mean Earnings Slope	0.073	0.058	0.057	
Labor Market Entry < 5 Years	0.156	0.140	0.126	
Pre-Start 0 Earnings Quarter	0.191	0.197	0.182	

This table describes key study variables in our unweighted and weighted samples, including a comparison to our work organization dimension rates to the BG data overall (after the processing described above). We created this table using data from Burning Glass, the U.S. Census Longitudinal Employer-Dynamics Data, the American Community Survey, and IPUMS public-use data.

Table A.3: How Work Organization Affects Starting Pay, Descriptive Results

Sample	A. Main					
Model	(1)	(2)	(3)	(4)	(5)	(6)
Complexity, Off	1.56***	0.472***	0.450***	0.254***	0.243***	0.125***
	(0.034)	(0.018)	(0.025)	(0.016)	(0.018)	(0.012)
Complexity, On	1.65***	0.576***	0.344***	0.253***	0.193***	0.155***
	(0.043)	(0.028)	(0.033)	(0.019)	(0.027)	(0.015)
Discretion	0.750***	0.069*	0.145***	0.107***	0.071**	0.065***
	(0.056)	(0.029)	(0.033)	(0.021)	(0.025)	(0.016)
Relationships	-0.350***	-0.330***	-0.105***	-0.022	-0.088***	-0.025*
	(0.043)	(0.025)	(0.025)	(0.016)	(0.019)	(0.012)
Start Quarter	×	×	×	×	×	×
SOC5			×	×	×	×
Naics5 + Size				×		×
Human Capital		×			×	×
R^2	0.20	0.63	0.47	0.54	0.67	0.70
Within R^2	0.18	0.62	0.01	0.00	0.40	0.36

Sample	B. Not Managerial/Professional					
Model	(7)	(8)	(9)	(10)	(11)	(12)
Complexity, Off	0.933***	0.459***	0.490***	0.210***	0.276***	0.105***
	(0.047)	(0.028)	(0.037)	(0.022)	(0.028)	(0.017)
Complexity, On	1.25***	0.469***	0.303***	0.237***	0.166***	0.144***
	(0.045)	(0.031)	(0.039)	(0.021)	(0.030)	(0.016)
Discretion	0.404***	0.090*	0.102*	0.105***	0.055	0.070***
	(0.069)	(0.040)	(0.043)	(0.026)	(0.032)	(0.019)
Relationships	-0.561***	-0.344***	-0.151***	-0.023	-0.111***	-0.017
	(0.046)	(0.029)	(0.031)	(0.019)	(0.024)	(0.014)
Start Quarter	×	×	×	×	×	×
SOC5			×	×	×	×
Naics5 + Size				×		×
Human Capital		×			×	×
R^2	0.17	0.55	0.38	0.49	0.60	0.65
Within R^2	0.14	0.53	0.02	0.01	0.36	0.32

Significance: * $p < .05$; ** $p < .01$; *** $p < .001$ (two-tailed tests). The dependent variable is second quarter pay throughout. The main sample (Panel A) comprises 611,000 job starts by 570,000 workers at 23,500 employers and the non-managerial/professional sample (Panel B) comprises 293,000 job starts by 277,000 workers at 15,000 employers. We created this table using data from Burning Glass and the U.S. Census Longitudinal Employer-Dynamics Data linked to the American Community Survey.

Table A.4: How Work Organization Affects Starting Pay, Managerial/Professional Sample

Model	(1)	(2)	(3)	(4)	(5)
Complexity, Off	0.850***	0.483***	0.175***	0.140***	0.076***
	(0.048)	(0.029)	(0.024)	(0.0223)	(0.017)
Complexity, On	2.09***	0.533***	0.147***	0.123***	0.077**
	(0.082)	(0.044)	(0.037)	(0.035)	(0.026)
Discretion	0.258***	0.263***	0.108**	0.098**	0.083***
	(0.056)	(0.040)	(0.033)	(0.031)	(0.023)
Relationships	-0.607***	0.034	0.027	0.032	0.008
	(0.048)	(0.030)	(0.027)	(0.025)	(0.019)
Start Quarter	×	×	×		
SOC5		×			
SOC5∧Company			×	×	×
StartQ∧CZ∧SOC2				×	×
Human Capital					×
R^2	0.07	0.32	0.55	0.62	0.77
Within R^2	0.05	0.01	0.00	0.00	0.39

Significance: * $p < .05$; ** $p < .01$; *** $p < .001$ (two-tailed tests). The dependent variable is second quarter pay throughout. This managerial/professional sample comprises 317,000 job starts by 294,000 workers at 14,500 employers. We created this table using data from Burning Glass and the U.S. Census Longitudinal Employer-Dynamics Data linked to the American Community Survey.

Table A.5: How Work Organization Affects Starting Pay, Results Preferring Same Quarter of Job Post and Job Start

Sample	Main Sample		Man./Prof.		Not Man./Prof.	
Model	(1)	(2)	(3)	(4)	(5)	(6)
Complexity, Off	0.102***	0.053***	0.137***	0.074***	0.080***	0.031*
	(0.013)	(0.011)	(0.022)	(0.017)	(0.019)	(0.016)
Complexity, On	0.099***	0.070***	0.120***	0.068**	0.101***	0.075***
	(0.014)	(0.012)	(0.035)	(0.025)	(0.016)	(0.014)
Discretion	0.067***	0.058***	0.075*	0.058*	0.069***	0.060***
	(0.016)	(0.013)	(0.031)	(0.023)	(0.018)	(0.016)
Relationships	0.018	0.015	0.051*	0.031	0.008	0.011
	(0.013)	(0.011)	(0.025)	(0.019)	(0.015)	(0.013)
SOC5∧Compan×	×	×	×	×	×	×
StartQ∧CZ∧SOC2	×	×	×	×	×	×
Human Capital		×		×		×
R^2	0.73	0.82	0.62	0.77	0.71	0.79
Within R^2	0.00	0.33	0.00	0.39	0.00	0.27

Significance: * $p < .05$; ** $p < .01$; *** $p < .001$ (two-tailed tests). The dependent variable is second quarter pay throughout. This 'main' sample comprises 611,000 job starts by 570,000 workers at 23,500 employers. Rather than preferring job posts in in the quarter prior to job start, as in our main results, we here prefer job posts in the same quarter as the job start. We created this table using data from Burning Glass and the U.S. Census Longitudinal Employer-Dynamics Data linked to the American Community Survey.

Table A.6: Characteristics of Employers Offering Complex and Autonomous Front-Line Tasks, Residualized by 2-Digit NAICS Industry

Variable	Sample Means	On	Off	Discretion	Relationships
Employment	1150	461	454	-423	194
Logged Employment	5.0	0.2	0.4	0.0	0.0
Logged Payroll	8.8	0.2	0.6	0.0	0.0
Logged Revenue	9.4	0.3	0.5	0.0	0.0
Employment Growth	0.05	-0.02	0.00	0.02	0.00
Revenue Growth	0.27	-0.01	0.01	0.02	0.02
Revenue/Employee	162.0	11.9	36.9	7.1	-10.5
Profits/Employee	106.0	10.8	25.9	0.9	-9.8
Pay/Employee	55.9	1.2	10.9	6.1	-0.7
Mean ln Q2 Pay	8.96	0.05	0.15	0.05	0.00
Variance lnQ2 Pay	0.23	0.01	0.01	0.00	0.01
Founded Pre-1980	0.29	0.03	-0.01	-0.03	-0.01
Founded 1980-1999	0.31	-0.01	0.01	0.00	-0.01
Founded Post-2000	0.40	-0.02	0.00	0.03	0.02
C Corporation	0.29	0.04	0.05	-0.01	0.00
S-Corp/Partnership	0.56	-0.02	-0.10	-0.02	0.01
Non-Profit	0.15	-0.02	0.05	0.01	-0.01

The sample used in this table comprises 59,000 employers. Columns labels with our work organization scores reflect the difference in average value between employers that are above versus below the median of the focal work organization dimension. 'Off' and 'On' refer to our off-the-job complexity and on-the-job complexity measures, respectively. The predictors were created using Burning Glass data. We residualize all values by 2-digit Naics, but otherwise create the table in the same way as our employers-level results in the main text. Mean second quarter pay and variance in second quarter pay, were created using the U.S. Census Longitudinal Employer-Dynamics Database; and all other variables were created using the Census Longitudinal Business Database.

Figure A.1: Complexity Survey Instructions

We are studying how people interpret job requirements that are commonly included in online job postings. Your participation will help us give advice about how to write job postings that can attract successful applicants.

To do this, we will show you a series of job requirements that employers list on job postings. Imagine someone you know was applying to a position between 2010 to 2018, and one of the following job requirements was listed. Choose the category that you think best fits that requirement:

> (1) **Simple**: by simple, we mean the requirement implies a relatively straightforward task that most people could do without advanced education or substantial prior work experience. For example, "line cook" or "cash handling" would both be simple requirements.
>
> (2) **Complex, learned on the job**: this category means the job requirement has a **more complex skill set**, but can typically be **learned on the job**. For example, "electrical wiring" and "office management" both typically require some practice and on-the-job learning before an employee is proficient at them.
>
> (3) **Complex, learned off the job**: job requirements in this group also require a **more complex skill set**, and must be learned **outside of the job**, such as in college. An example of this would be "computer programming" or "physical therapy", because these require a minimum of a bachelor's degree to be performed.

Before we move to the main survey, here is an example of the multiple choice questions you will see in the survey. We use this to gauge your understanding of the coding scheme.

	Simple	Complex, learned **on** the job	Complex, learned **off** the job	I don't know
Plumbing	○	○	○	○
Operating a cash register	○	○	○	○
Real estate sales	○	○	○	○
Teaching	○	○	○	○
Package delivery	○	○	○	○
Mental health therapy	○	○	○	○

Figure A.2: Autonomy Survey Instructions

We are studying how people interpret job requirements that are commonly included in online job postings. Your insight will help us give advice about how to write job postings that can attract successful applicants.

To do this, we will show you a series of job requirements that employers list on job postings. Imagine someone you know was applying to a position between 2010 to 2018, and one of the following job requirements was listed. Choose the category that you think best fits that requirement:

> (1) **High discretion**: By "high discretion", we mean that an employee has a high level of autonomy to decide what to do in their job. For example, "materials selection" or "strategic planning" imply high discretion, while "pipe systems" and "co-pay collection" do not. High discretion jobs tend to place responsibility on an employee and require decision-making. These jobs are difficult to supervise and assess, so managers need to trust employees to do their best and use their best judgment.
>
> (2) **Rule-bound or closely supervised**: By "rule-bound or closely supervised," we mean the job requirement implies either a high degree of supervision by a manager or supervisor or following detailed rules that limit employee decision-making. Unlike high discretion, this implies strong limits on employee autonomy and decision making. For example, jobs involving installation of "conveyor systems" or "reordering supplies" suggest high supervision and little decision making.
>
> (3) **Significant relationships with coworkers and clients**: In this category, we would like you to identify job requirements that centrally involve building important relationships with coworkers and clients. For example, "human resource management" and "financial advisor" require significant relationships with coworkers and clients to perform, while "electrical design" and "preparing reports" tend to be more autonomous. In jobs with significant relationships, employee autonomy and decision-making is constrained, not by rules, but by the need to navigate relationships with others.

Before we move to the main survey, here is an example of the multiple choice questions you will see in the survey. We use this to gauge your understanding of the coding scheme.

	High discretion	Rule-bound or closely supervised	Significant relationships with coworkers and clients	I don't know
Teaching	○	○	○	○
Mental health therapy	○	○	○	○
Shelf restocking	○	○	○	○
Blog posts	○	○	○	○
Package delivery	○	○	○	○
Real estate sales	○	○	○	○